

Reed–Solomon List Decoding: Certificates, Refinements, and Obstructions

Quang Dao*

Scott Duke Kominers[†]

Justin Thaler[‡]

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Abstract

Hidden-derivative interpolation for Reed–Solomon codes raises quantitative questions beyond the existence of an all-rate decoder: how to compute finite certificates, how characteristic affects coefficient lifting, and how large agreement lists can be near capacity. This companion collects exact rank formulas, differential-root refinements, and combinatorial constructions addressing these questions. We give finite block decompositions of the local interpolation map, fixed-parameter certificates of orders zero and one, and a fixed-gap boundary asymptotic for a one-sided support. We also analyze descent to the data field, resonant Taylor coefficients, and randomized regular witnesses. Coefficient fibers and multiplicative cosets give large agreement lists, while a Frobenius-plane construction obstructs enumeration from higher-derivative-supported equations in fixed characteristic. The all-rate decoding theorem and the geometric list-size and correlated agreement results are developed in the separate main paper.

1 Introduction

Our main paper, *Reed–Solomon Codes Beyond Johnson: Efficient Decoding and Smaller Cryptographic Proofs*, develops explicit list-decoding and agreement bounds from ordinary and hidden-derivative interpolation, together with a deterministic decoder. This companion collects the auxiliary results that support further study of the method. Its purpose is to retain the finite formulas, alternative root arguments, and obstructions without reproducing the main decoding theorem or assembling its proof. The two manuscripts have separate introductions and result statements, and share basic definitions and the differential-root core.

Finite interpolation. [Section 4](#) gives the exact dimension of a derivative- weighted monomial space, the kernel of the full local map, an exact rank for an enlarged map, and a block decomposition of the actual map. [Section 5](#) preserves the one-sided dimension and coordinate estimates behind the earlier support analysis. These formulas distinguish an exact rank from a coordinate-count upper bound and yield a finite interpolation certificate. [Section 6](#) preserves fixed-parameter low-order examples and the logarithmically shrinking-gap consequence of the main paper’s certificate. [Section 7](#) preserves finite first-order and Johnson constants, exact Johnson tuning, and the proved dominance over an earlier full-differentiation envelope. [Section C](#) analyzes a different, one-sided support with the rate and gap held fixed as the derivative order grows.

*Carnegie Mellon University. qvd@andrew.cmu.edu.

[†]Harvard University and a16z crypto research. kominers@fas.harvard.edu.

[‡]a16z crypto research and Georgetown University. jthaler@a16z.com.

Root finding and characteristic. Section 7.4 records the regular-lifting and separant-chain lemmas used by the main paper, then develops alternative all-root and stratified bounds. Sections 7.8 and 8 treat primitive normalization, descent to the field generated by the data, resonant coefficient lifting, and randomized regular witnesses. The extension to characteristic growing linearly with the block length is a corollary of the main paper’s interpolation certificate and the resonant root bound. The Frobenius obstruction in Section 9 concerns the present support and its unfiltered differential roots, not every possible list decoder.

Agreement lists near capacity. Section 10 gives elementary upper bounds, coefficient-fiber lower bounds for arbitrary evaluation sets, and multiplicative-coset constructions. The distinctions between fixed gap, varying gap, and sufficiently large block length are part of the statements. These examples constrain list-size guarantees without giving matching lower bounds on the derivative order of hidden-derivative interpolation.

Reading the companion. We recall the polynomial and local-interpolation conventions, so most auxiliary arguments can be read here directly. References explicitly marked “of the main paper” link to [the accompanying main PDF](#); their numbering belongs to that manuscript. In particular, the support certificate is an external input where cited. The exact counting and source-audit appendices are retained here. The main paper’s verification paragraph describes the checked Lean scope; it does not assert formal verification of all the auxiliary results collected in this companion.

Authorship and assistance. This companion was separated from the same AI-assisted research draft as the main paper. Its account of assistance and author responsibility applies to both manuscripts. The authors are responsible for all statements and proofs retained here.

2 Preliminaries

We collect the notation and background used by the interpolation, root-counting, and correlated-agreement arguments. The discussion fixes our agreement and Hasse-derivative conventions, records the backward Taylor identity, and introduces the Taylor-jet and algebraic-geometric language needed later.

Throughout the paper, $\mathbb{F}_q$ denotes a finite field, $\mathbb{N} = \{0, 1, 2, \dots\}$, and $[n] = \{1, \dots, n\}$. The local interpolation argument in Section 3 works in arbitrary characteristic; root-finding statements specify their characteristic hypotheses. We write $\mathbb{F}_q[X]_{\leq D}$ for the vector space of univariate polynomials of degree at most D , including the zero polynomial. We take $\deg 0 = -\infty$, so the degree inequalities below also include it.

Counting and probability notation. For a real number x , its positive part is $x_+ := \max\{x, 0\}$. Thus $(u - z)_+^2$ means $(\max\{u - z, 0\})^2$; in the integrals below the integrand is zero whenever $z \geq u$. The indicator $\mathbf{1}_{\{\mathcal{P}\}}$ equals one when the condition $\mathcal{P}$ holds and zero otherwise. We write $\#S$ or $|S|$ for the cardinality of a finite set S . For a nonnegative vector x , $|x|$ denotes the sum of its coordinates; for a scalar it denotes absolute value.

We write $\Pr[\mathcal{A}]$ for the probability of an event and $\mathbb{E}[U]$ for the expectation of a random variable U , under the specified distribution. In particular, $\mathbb{E}[U^2]$ is the second moment, whereas $(\mathbb{E}[U])^2$ is the square of the mean. The variance is $\text{Var}(U) = \mathbb{E}[(U - \mathbb{E}[U])^2]$. For a Euclidean region S , $\text{vol}(S)$ denotes its Lebesgue volume; a uniform point of S is sampled from normalized volume.

Unless a base is displayed, $\log$ denotes the natural logarithm. We use the harmonic sums $H_t = \sum_{j=1}^t j^{-1}$ and $H_t^{(2)} = \sum_{j=1}^t j^{-2}$, with both sums zero at $t = 0$. A subscript on asymptotic notation, such as $O_\delta(\cdot)$, indicates that its implicit constant may depend on that parameter. Section-specific notation is introduced where its role in the argument becomes relevant.

2.1 Reed–Solomon codes and agreement

Fix pairwise distinct evaluation points $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{F}_q^n$. The Reed–Solomon code of message dimension k on this evaluation set is

$$\text{RS}_\alpha[n, k] := \{(P(\alpha_1), \dots, P(\alpha_n)) : P \in \mathbb{F}_q[X], \deg P < k\}. \quad (1)$$

Thus the code has rate $\rho = k/n$; no multiplicative or additive structure is assumed of the evaluation set.

For $u, v \in \mathbb{F}_q^n$, their agreement is the number of coordinates on which they coincide:

$$\text{agr}(u, v) := |\{i \in [n] : u_i = v_i\}|. \quad (2)$$

Given a received word $y \in \mathbb{F}_q^n$ and an integer A , the agreement list at ambient degree D is

$$\mathcal{L}_D(y, A) := \{P \in \mathbb{F}_q[X]_{\leq D} : \text{agr}((P(\alpha_i))_{i=1}^n, y) \geq A\}. \quad (3)$$

The decoder will first work in such an ambient space and then retain only the polynomials of degree less than the actual message dimension.

2.2 Hasse derivatives and multiplicity

We use the standard Hasse-derivative convention [DKSS13].

Definition 2.1 (Hasse derivative). For $P \in \mathbb{F}_q[X]$, its j -th Hasse derivative $P^{[j]}$ is defined by the polynomial identity

$$P(X + T) = \sum_{j \geq 0} P^{[j]}(X) T^j. \quad (4)$$

Equivalently, if $P(X) = \sum_i p_i X^i$, then

$$P^{[j]}(X) = \sum_{i \geq j} \binom{i}{j} p_i X^{i-j}. \quad (5)$$

The definition is valid in every characteristic and gives $\deg P^{[j]} \leq D - j$ whenever $\deg P \leq D$, with a zero derivative when $j > D$.

Ordinary Taylor expansion uses derivatives at X to obtain the value at $X + T$. The backward identity below uses derivatives at $X + T$ to recover the value at X . The hidden-derivative constraints use this inversion of Equation (4). We include the short proof because every sign and power of T enters the local multiplicity calculation.

Lemma 2.2 (Backward Taylor identity). *Every $P \in \mathbb{F}_q[X]$ satisfies*

$$P(X) = \sum_{j \geq 0} (-T)^j P^{[j]}(X + T) \quad \text{in } \mathbb{F}_q[X, T]. \quad (6)$$

Consequently, for every $d \geq 0$,

$$P(X + T) \equiv P(X) + \sum_{j=1}^d (-1)^{j+1} T^j P^{[j]}(X + T) \pmod{T^{d+1}}. \quad (7)$$

Proof. Apply Equation (4) with X replaced by $X + T$ and the increment replaced by $-T$. This gives Equation (6). Isolating its $j = 0$ term and discarding the terms with $j > d$ gives Equation (7). $\square$

Definition 2.3 (Multiplicity). For $F \in \mathbb{F}_q[X]$ and $\alpha \in \mathbb{F}_q$, the multiplicity of F at α is

$$\text{mult}_\alpha(F) := \max\{s \in \mathbb{N} : T^s \text{ divides } F(\alpha + T)\}. \quad (8)$$

We set $\text{mult}_\alpha(0) = \infty$.

We use the standard fact that a nonzero univariate polynomial cannot have more roots, counted with multiplicity, than its degree. In the form needed below, if $\alpha_1, \dots, \alpha_t$ are distinct and $\text{mult}_{\alpha_i}(F) \geq m$ for every i , then either $F = 0$ or $\deg F \geq mt$.

For a bivariate polynomial $Q(X, Y)$, multiplicity at least s at (α, y) means that every coefficient of $U^a V^b$ with $a + b < s$ vanishes in $Q(\alpha + U, y + V)$. This is the notion used by the order-zero interpolation argument. If $P(\alpha) = y$, both T and $P(\alpha + T) - y$ are divisible by T . Substituting them for U, V therefore shows that $Q(X, P(X))$ has a root of multiplicity at least s at $X = \alpha$.

2.3 Taylor data and polynomial differential equations

Taylor coefficients and jets. At a chosen point a , the identity $P(a + T) = \sum_j P^{[j]}(a)T^j$ writes the polynomial in powers of $X - a$. We call a the *Taylor center* and the values $P^{[j]}(a)$ its *Taylor coefficients* there. An order- r *jet* is simply the finite tuple

$$(P(a), P^{[1]}(a), \dots, P^{[r]}(a)).$$

Equivalently, it records $P(a + T)$ modulo T^{r+1} . Thus “initial Taylor data” and “initial jet” refer to these first $r + 1$ coefficients; neither term requires analytic power series or convergence.

Differential equations and jet degree. We encode a polynomial differential equation by a nonzero polynomial $Q(X, Y_0, \dots, Y_r)$. A polynomial P solves it when substituting $Y_j = P^{[j]}(X)$ makes Q identically zero. The Y_j are called *jet variables* because, at a fixed $X = a$, they stand for the entries of the jet. The *order* is the largest index r on which Q actually depends. Its *total jet degree* is its total degree in $Y_0, \dots, Y_r$, ignoring X and any challenge variable. These are different quantities: $Q = Y_1 - Y_0^2$ has order one and jet degree two.

The separant and regular Taylor data. For an equation of actual order r , its *separant* is the ordinary partial derivative

$$S = \frac{\partial Q}{\partial Y_r}.$$

A solution’s jet at a is *regular for this equation* if $S(a, P(a), \dots, P^{[r]}(a)) \neq 0$; otherwise we call it singular. The separant matters because it multiplies each new unknown coefficient in the reconstruction equation. Under the characteristic hypotheses used below, a regular jet therefore determines successive coefficients uniquely. Singular solutions are handled by passing to the separant equation and repeating; this is the *separant chain*.

Taylor reconstruction and charts. A *Taylor recurrence* is the equation used to compute the next coefficient from those already known. A *lift* extends an initial jet to a longer coefficient sequence satisfying the equation to the required order. The resulting formulas express the later coefficients as rational functions of the initial ones. We call this parameterization a *Taylor chart*, restricted to the initial data where its denominators are nonzero. In particular, a chart is a map from a short tuple of coefficients to the reconstructed polynomial coefficients; its degree bounds control the subsequent counting argument.

2.4 Weighted degree

Definition 2.4 (Weighted degree). For nonnegative integer weights $w_0, \dots, w_s$, the weighted degree of a monomial $Z_0^{e_0} \cdots Z_s^{e_s}$ is $\sum_{i=0}^s w_i e_i$. The weighted degree $\text{wdeg}_{w_0, \dots, w_s} Q$ of a nonzero polynomial Q is the maximum weighted degree of one of its monomials with nonzero coefficient. We set $\text{wdeg}_{w_0, \dots, w_s}(0) = -\infty$.

The weights relevant to polynomial differential equations record the actual degree lost under Hasse differentiation. Namely, if $d \leq D$, $Q \in \mathbb{F}_q[X, Y_0, \dots, Y_d]$, and $P \in \mathbb{F}_q[X]_{\leq D}$, then

$$\deg_X Q(X, P(X), P^{[1]}(X), \dots, P^{[d]}(X)) \leq \text{wdeg}_{1, D, D-1, \dots, D-d} Q. \quad (9)$$

Indeed, the variable X contributes degree one and the substitution for Y_j contributes degree at most $D - j$. This elementary accounting is the reason to distinguish the actual derivative weights from the common upper bound D used in the weighted-support certificate.

2.5 Terminology for counting polynomial families

The counting proofs use sets of coefficient tuples satisfying polynomial equations. We recall the geometric terminology here; the specific counting inequalities are stated where they are used.

An *affine algebraic set* is a common zero set of polynomials in finitely many coordinates. We work over an algebraically closed field, meaning that every nonconstant univariate polynomial splits into linear factors. A *locus* simply means the set satisfying the stated conditions; these may include inequalities such as $S \neq 0$. A *hypersurface* is the zero set of one nonconstant polynomial; a *hyperplane* is the case of a degree-one equation. The *Zariski closure* of a set is the smallest algebraic set containing it. An *irreducible component* is a maximal piece that cannot be written as a union of two proper algebraic subsets.

Intuitively, dimension counts the independent parameters in such a family. The *degree* of an irreducible s -dimensional algebraic set is the number of points, counted with multiplicity, obtained by intersecting its projective closure (which also includes points at infinity) with a linear subspace of codimension s in general position. Here general position means that the choices avoid the algebraic conditions causing a degenerate intersection. A point and a linear space each have degree one. For a set with several components, our *total degree* sums their degrees, including components of different dimensions. The degree bounds limit how many pieces can remain as we impose agreement equations.

A *rational map* has coordinates that are ratios of polynomials; it is used only where its denominators are nonzero. Its *image closure* is the Zariski closure of the values it takes. An *incidence* is a point together with an equation it satisfies. Thus counting agreement incidences means counting pairs consisting of a candidate and a coordinate where it agrees with the received word.

A staircase count. For the finite certificates and boundary asymptotics, we use

$$G(z) = \sum_{j \geq 0} (z - j)_+, \quad (t)_+ = \max\{t, 0\}. \quad (10)$$

Only finitely many summands are nonzero. The count $N_D(L) = \#\{(u, a) \in \mathbb{N}^2 : u + Da < L\}$ satisfies $N_D(L) = DG(L/D)$ for integer L and $D \geq 1$.

3 Hidden-derivative interpolation

We use hidden-derivative interpolation to construct a nonzero equation satisfied by every sufficiently close message. We call the polynomial representing such an equation a hidden-derivative explainer, or simply an explainer.

Definition 3.1 (Explainer). Fix a field F , distinct evaluation points $\alpha_1, \dots, \alpha_n \in F$, a received word $w \in F^n$, integers $K \geq 1$ and $1 \leq A \leq n$, and an order bound $d \geq 0$. A nonzero polynomial $Q \in F[X, Y_0, \dots, Y_d]$ is an *explainer* for w at agreement threshold A and degree bound K if every $P \in F[X]$ with $\deg P < K$ satisfies

$$|\{i : P(\alpha_i) = w_i\}| \geq A \implies Q(X, P, P^{[1]}, \dots, P^{[d]}) \equiv 0.$$

The implication is one-way: an explainer can have polynomial solutions that are far from the received word. Agreement must still be imposed when recovering or counting the requested messages. The ambient degree bound K is part of the definition and may exceed the actual message dimension k ; explaining all degree- $< K$ candidates also explains the degree- $< k$ candidates.

3.1 Local interpolation constraints

BCPZZ's backward-Taylor substitution turns agreement into multiplicity without requiring derivative evaluations [BCP⁺26, Section 3.1]. Fix an order $d \geq 0$ and a target multiplicity $m \geq 1$. For formal variables $Y = (Y_1, \dots, Y_d)$, write $B_d(T, Y) = \sum_{j=1}^d (-1)^{j+1} T^j Y_j$; an empty sum is zero. The identity in Equation (7) says that, at an agreement, the remainder after these d terms is divisible by T^{d+1} . We encode that divisibility directly in the local substitution.

Definition 3.2 (Local hidden-derivative constraints). For $(\alpha, y) \in \mathbb{F}^2$ and $Q \in \mathbb{F}[X, Y_0, \dots, Y_d]$, put

$$\Phi_{\alpha, y}^{d, m}(Q) := Q(\alpha + T, y + B_d(T, Y) + T^{d+1}V, Y_1, \dots, Y_d) \pmod{T^m} \quad (11)$$

in $\mathbb{F}[T, V, Y_1, \dots, Y_d]/(T^m)$. The multiplicity- m , order- d constraints at (α, y) are

$$\Phi_{\alpha, y}^{d, m}(Q) = 0. \quad (12)$$

These are homogeneous linear conditions on any fixed interpolation space. They require every coefficient in $V, Y_1, \dots, Y_d$ to be divisible by T^m before reduction.

Lemma 3.3 (Local multiplicity [BCP⁺26, Lemma 3.1]). *If $P(\alpha) = y$ and Q satisfies (12), then*

$$Q(\alpha + T, P(\alpha + T), P^{[1]}(\alpha + T), \dots, P^{[d]}(\alpha + T)) \equiv 0 \pmod{T^m}. \quad (13)$$

Thus each agreement gives a root of multiplicity at least m after substituting the candidate and its derivatives into Q .

Proof. The backward Taylor identity makes

$$V_P(T) = \frac{P(\alpha + T) - y - \sum_{j=1}^d (-1)^{j+1} T^j P^{[j]}(\alpha + T)}{T^{d+1}} \quad (14)$$

a polynomial. Substitute $Y_j = P^{[j]}(\alpha + T)$ and $V = V_P(T)$ into (12). The second input to Q becomes $P(\alpha + T)$, proving the claim in every characteristic. $\square$

For the rank calculations we retain the equivalent remainder variable $E = T^d V$. Substitute $Y_0 = y + B_d(T, Y) + TE$ and retain the rows

$$T^i E^u Y^j \quad \text{with } i + du < m. \quad (15)$$

Rows with $i < u$ are structurally zero, since each E comes with a factor of T . On the remaining rows, the substitution relabels

$$(i, u, \mathbf{j}) \mapsto (i + du, u, \mathbf{j}). \quad (16)$$

Its image is exactly the possible nonzero rows of (11), whose T -degrees are at least $(d+1)u$ and below m . Thus it preserves every matrix entry, kernel, rank, and jet-degree grouping. If y depends on a challenge, its challenge degrees are preserved as well. We can therefore compute ranks in the E coordinates while using Lemma 3.3 for every order.

3.2 Obtaining an explainer from the local constraints

We now combine the conditions from all received positions. Enough available coefficients will guarantee a nonzero interpolant; enough agreements will force its specialization to vanish identically. The result is an explainer for the received word. The following principle separates these requirements so that we can next choose a support satisfying them.

Let $\mathcal{Q} \subseteq \mathbb{F}_q[X, Y_0, \dots, Y_d]$ be a finite-dimensional monomial space. The local constraints define a linear map $\Phi_{\alpha, y}$ from $\mathcal{Q}$ to the vector of coefficients required to vanish in Equation (12). Write rank_{loc} for a common upper bound on $\text{rank}(\Phi_{\alpha, y})$, uniform over (α, y) .

The next proposition abstracts the final dimension-versus-rank and multiplicity step of [BCP⁺26, Proposition 3.13]. The ambient-degree formulation distinguishes the message dimension from the larger space used for interpolation; the underlying argument is theirs.

Proposition 3.4 (BCPZZ global interpolation principle, in ambient-degree form). *Fix distinct $\alpha_1, \dots, \alpha_n \in \mathbb{F}_q$, a received word $y = (y_1, \dots, y_n)$, and an agreement threshold A . Suppose that:*

(i) *every $Q \in \mathcal{Q}$ and $P \in \mathbb{F}_q[X]_{\leq D}$ satisfy*

$$\deg_X Q(X, P, P^{[1]}, \dots, P^{[d]}) < mA; \quad (17)$$

(ii) *the local constraint map at each received point has rank at most rank_{loc} ; and*

(iii) $\dim \mathcal{Q} > n \text{rank}_{\text{loc}}$.

Then there is a nonzero $Q \in \mathcal{Q}$ satisfying the local constraints at every (α_i, y_i) . Every polynomial $P \in \mathbb{F}_q[X]_{\leq D}$ that agrees with y in at least A positions then satisfies the polynomial identity

$$Q(X, P(X), P^{[1]}(X), \dots, P^{[d]}(X)) \equiv 0. \quad (18)$$

Proof. Stack the n local maps into one homogeneous linear map on $\mathcal{Q}$. Its rank is at most $n \text{rank}_{\text{loc}}$, so the strict dimension inequality gives a nonzero element Q in its kernel.

Fix a polynomial P with at least A agreement positions and set

$$F_P(X) := Q(X, P(X), P^{[1]}(X), \dots, P^{[d]}(X)). \quad (19)$$

At each agreement position, Lemma 3.3 gives a root of F_P of multiplicity at least m . The roots are distinct, so a nonzero F_P would have degree at least mA . This contradicts Equation (17); hence $F_P = 0$. $\square$

4 Exact finite interpolation refinements

We now choose an interpolation space and evaluate both sides of the dimension versus rank comparison without asymptotic estimates. BCPZZ introduce the weighted derivative support, factor the local map through an enlarged map Γ , and exhibit independent subspaces of $\ker \Gamma$. We retain these objects but strengthen their analysis in four ways. Each derivative variable is charged its actual specialization degree, no separate upper bound is placed on the ordinary degree in $Y_2, \dots, Y_d$, the exhibited kernel subspaces are proved to exhaust $\ker \Gamma$, and the actual local map is decomposed into exact finite blocks. The resulting test is a finite inequality in integers.

Fix integers

$$1 \leq A \leq n, \quad 1 \leq d < D, \quad m \geq 1, \quad M \geq 0, \quad W \geq 0. \quad (20)$$

For $c = (c_2, \dots, c_d) \in \mathbb{N}^{d-1}$, define

$$|c| := \sum_{j=2}^d c_j, \quad \omega(c) := \sum_{j=2}^d (j-1)c_j, \quad Y^c := \prod_{j=2}^d Y_j^{c_j}. \quad (21)$$

We use the weighted-simplex count

$$\Lambda_d(z) := |\{c \in \mathbb{N}^{d-1} : \omega(c) \leq z\}|, \quad (22)$$

with the convention $\Lambda_d(z) = 0$ for $z < 0$.

4.1 The derivative-weighted interpolation space

Definition 4.1 (Derivative-weighted interpolation space). Let $Q(D, A, d, m, M, W)$ be the span of all monomials $X^a Y_0^{b_0} Y_1^{b_1} Y^c$ indexed by nonnegative integers a, b_0, b_1 and $c \in \mathbb{N}^{d-1}$ such that

$$b_1 \leq M, \quad \omega(c) \leq W, \quad (23)$$

$$a + Db_0 + (D-1)b_1 + \sum_{j=2}^d (D-j)c_j < mA. \quad (24)$$

Because $D-j > 0$ for every $j \leq d$, Equation (24) makes this a finite-dimensional space even without a separate bound on $|c|$.

Lemma 4.2 (Degree bounds). *Every $Q \in Q(D, A, d, m, M, W)$ has the following properties:*

$$\text{wdeg}_{1,D,D-1,\dots,D-d} Q < mA, \quad (25)$$

$$\deg_X Q(X, P, P^{[1]}, \dots, P^{[d]}) < mA \quad \text{for every } P \in \mathbb{F}_q[X]_{\leq D}, \quad (26)$$

$$\deg_{Y_0,\dots,Y_d} Q \leq \left\lfloor \frac{mA-1}{D-d} \right\rfloor. \quad (27)$$

Proof. The first claim is the defining inequality. The second follows from Equation (9). For the last claim, every Y_j has weight at least $D-d$. A monomial of total Y -degree s therefore has weighted degree at least $(D-d)s$; the strict integer inequality in Equation (24) gives $(D-d)s \leq mA-1$. $\square$

To state the dimension exactly, define the staircase count

$$N_D(t) := \sum_{b \geq 0} (t - Db)_+ = \begin{cases} \sum_{b=0}^{\lfloor (t-1)/D \rfloor} (t - Db), & t > 0, \\ 0, & t \leq 0. \end{cases} \quad (28)$$

For integer t , this is the number of pairs $(a, b) \in \mathbb{N}^2$ satisfying $a + Db < t$.

Proposition 4.3 (Exact dimension of the interpolation space). *The interpolation space in Definition 4.1 has dimension*

$$\dim Q(D, A, d, m, M, W) = \sum_{\substack{c \in \mathbb{N}^{d-1} \\ \omega(c) \leq W}} \sum_{b_1=0}^M N_D \left(mA - (D-1)b_1 - \sum_{j=2}^d (D-j)c_j \right). \quad (29)$$

Proof. Fix c and b_1 satisfying Equation (23). The remaining indices (a, b_0) satisfy

$$a + Db_0 < mA - (D-1)b_1 - \sum_{j=2}^d (D-j)c_j. \quad (30)$$

By Equation (28), their number is the corresponding summand in Equation (29). Distinct index tuples give distinct monomials, so summing over c and b_1 gives equality. $\square$

4.2 The full local kernel

Before passing to the enlarged map, it is useful to record the exact kernel of the normalized local substitution on the full polynomial ring. This description is independent of any interpolation support and is new to this work. Put

$$B := Y_0 - \sum_{j=1}^d (-1)^{j+1} X^j Y_j. \quad (31)$$

After normalizing the received point to $(0, 0)$ and renaming the direct remainder variable V from Equation (11) as Z , the local map is

$$X \mapsto T, \quad B \mapsto T^{d+1}Z, \quad Y_j \mapsto Y_j \quad (1 \leq j \leq d), \quad (32)$$

with the target reduced modulo T^m . Denote this full-ring map by $\Phi_{0,0}^{\text{full}}$.

Proposition 4.4 (Exact kernel of the full local map). *The kernel of (32) is the monomial ideal in the coordinates $(X, B, Y_1, \dots, Y_d)$*

$$\mathcal{I}_m = \left(X^{\max\{0, m-(d+1)b\}} B^b : 0 \leq b \leq \left\lceil \frac{m}{d+1} \right\rceil \right). \quad (33)$$

Consequently, for every finite-dimensional polynomial space $\mathcal{U}$,

$$\text{rank}(\Phi_{0,0}^{\text{full}}|_{\mathcal{U}}) = \dim \mathcal{U} - \dim(\mathcal{U} \cap \mathcal{I}_m). \quad (34)$$

At a general received point (α, y) , the same statement holds after replacing X by $X - \alpha$ and B by

$$B_{\alpha,y} := Y_0 - y - \sum_{j=1}^d (-1)^{j+1} (X - \alpha)^j Y_j.$$

Proof. The change of coordinates from Y_0 to B is triangular. Hence the monomials $X^a B^b Y_1^{c_1} \dots Y_d^{c_d}$ form a basis of the source. Under (32), such a monomial becomes

$$T^{a+(d+1)b} Z^b Y_1^{c_1} \dots Y_d^{c_d}.$$

Distinct source monomials that survive modulo T^m have distinct target monomials. The kernel is therefore spanned by those with $a + (d+1)b \geq m$, and the displayed generators are exactly the minimal monomials with that property. Restricting a linear map to $\mathcal{U}$ gives (34). Translation gives the final statement. $\square$

4.3 The exact enlarged-map rank

Fix one received point (α, y) . Following [BCP⁺26], first make only the substitutions

$$X = \alpha + T, \quad Y_0 = y + TU, \quad (35)$$

and reduce modulo T^m . The resulting map $\Psi_{\alpha, y}$ takes $\mathcal{Q}(D, A, d, m, M, W)$ into the space

$$\mathcal{V} := \text{span} \left\{ T^r U^a Y_1^b Y^c : \begin{array}{l} 0 \leq r < m, 0 \leq a \leq r, 0 \leq b \leq M, \\ c \in \mathbb{N}^{d-1}, \omega(c) \leq W + r \end{array} \right\}. \quad (36)$$

The relaxed bound $\omega(c) \leq W + r$ contains the image and is chosen so that the kernel below remains inside $\mathcal{V}$.

To compute the rank, we return to the equivalent E coordinates from Equation (15). Solving $Y_0 = y + TU$ for E gives

$$E = U - \sum_{j=1}^d (-1)^{j+1} T^{j-1} Y_j. \quad (37)$$

Let Γ rewrite a polynomial in $\mathcal{V}$ in the variables $T, E, Y_1, \dots, Y_d$, equivalently by substituting $U = E + \sum_{j=1}^d (-1)^{j+1} T^{j-1} Y_j$, and then retain exactly the coefficients of monomials $T^i E^b Y_1^{e_1} \dots Y_d^{e_d}$ for which $i + db < m$. We identify its nonzero target rows with those of the direct remainder map through Equation (16). With this identification, the local constraint map factors as

$$\Phi_{\alpha, y} = \Gamma \circ \Psi_{\alpha, y}, \quad (38)$$

and hence $\text{rank}(\Phi_{\alpha, y}) \leq \text{rank}(\Gamma)$.

For $0 \leq r < m$, define the least remainder exponent that crosses the local multiplicity threshold by

$$h_r := \left\lceil \frac{m - r}{d} \right\rceil. \quad (39)$$

For each such r , expand E according to Equation (37) and define the subspace

$$\mathcal{K}_r := T^r E^{h_r} \cdot \text{span} \left\{ U^a Y_1^b Y^c : \begin{array}{l} 0 \leq a \leq r - h_r, 0 \leq b \leq M - h_r, \\ c \in \mathbb{N}^{d-1}, \omega(c) \leq W + r \end{array} \right\}, \quad (40)$$

where a negative upper bound makes the corresponding span zero. The subspaces $\mathcal{K}_r$ are the kernel subspaces constructed in [BCP⁺26, Lemma 3.11]. BCPZZ prove that their direct sum lies in $\ker \Gamma$ and use its dimension as a lower bound. The following result retains the positive parts that disappear in their coarser asymptotic analysis and strengthens the inclusion to an equality.

Lemma 4.5 (Exact rank of the enlarged map). *The kernel of Γ is*

$$\ker \Gamma = \bigoplus_{r=0}^{m-1} \mathcal{K}_r, \quad (41)$$

where the spaces $\mathcal{K}_r$ are defined in Equation (40). This identity holds in every characteristic. Consequently,

$$\text{rank}(\Gamma) = R_d(m, M, W), \quad (42)$$

where

$$R_d(m, M, W) := \sum_{r=0}^{m-1} \Lambda_d(W + r) ((r + 1)(M + 1) - (r - h_r + 1)_+ (M - h_r + 1)_+). \quad (43)$$

In particular, for every received point (α, y) ,

$$\text{rank}(\Phi_{\alpha, y}) \leq R_d(m, M, W). \quad (44)$$

Proof. All calculations take place in $\mathbb{F}_q[T, U, Y_1, \dots, Y_d]/(T^m)$. The monomial basis in Equation (36) gives

$$\dim \mathcal{V} = \sum_{r=0}^{m-1} (r+1)(M+1)\Lambda_d(W+r). \quad (45)$$

We first check that $\mathcal{K}_r \subseteq \mathcal{V}$. In a term of the expansion of E^{h_r} , suppose the variables Y_j , for $j \geq 2$, contribute the extra exponent vector e . They contribute $\ell = \omega(e)$ powers of T . The resulting monomial has T -degree $r + \ell$, U -degree at most r , Y_1 -degree at most M , and ω -weight at most $W + r + \ell$. It therefore lies in $\mathcal{V}$, or vanishes modulo T^m when $r + \ell \geq m$.

Next rewrite an element of $\mathcal{K}_r$ in the variables used by Γ . Every resulting monomial contains at least $T^r E^{h_r}$. Since $r + dh_r \geq m$, none of these monomials is retained by Γ . Thus $\mathcal{K}_r \subseteq \ker \Gamma$.

The dimension of this kernel subspace is

$$\dim \mathcal{K}_r = (r - h_r + 1)_+ (M - h_r + 1)_+ \Lambda_d(W + r). \quad (46)$$

To verify both this equality and independence across different r , take a nonzero relation among the $\mathcal{K}_r$ and choose its smallest T -degree $r_\star$. Modulo T , $E \equiv U - Y_1$, so the coefficient of $T^{r_\star}$ contributed by $\mathcal{K}_{r_\star}$ is $(U - Y_1)^{h_{r_\star}} g$ for a nonzero polynomial g . This coefficient is nonzero and cannot be cancelled by a space with larger index. The same observation shows that multiplication by $(U - Y_1)^{h_r}$ introduces no dependence within $\mathcal{K}_r$.

It follows that the direct sum of the $\mathcal{K}_r$'s lies in $\ker \Gamma$. We prove the reverse inclusion by successively removing the lowest nonzero T -coefficient. Write an arbitrary $F \in \mathcal{V}$ uniquely as

$$F = \sum_{r=0}^{m-1} T^r f_r(U, Y_1, \dots, Y_d), \quad (47)$$

where every monomial of f_r has U -degree at most r , Y_1 -degree at most M , and ω -weight at most $W + r$ in $Y_2, \dots, Y_d$.

Suppose that $F \in \ker \Gamma$ is nonzero, and let r be the least index for which $f_r \neq 0$. After rewriting

$$U = E + Y_1 + \sum_{j=2}^d (-1)^{j+1} T^{j-1} Y_j,$$

the coefficient of T^r is

$$f_r(E + Y_1, Y_1, \dots, Y_d).$$

The map Γ retains its coefficients of E^b exactly when $r + db < m$, or equivalently when $0 \leq b < h_r$. Since $F \in \ker \Gamma$, all of these coefficients vanish. Therefore

$$E^{h_r} \mid f_r(E + Y_1, Y_1, \dots, Y_d), \quad (U - Y_1)^{h_r} \mid f_r(U, Y_1, \dots, Y_d). \quad (48)$$

Write $f_r = (U - Y_1)^{h_r} g_r$. The divisor is monic of degree h_r when viewed separately as a polynomial in either U or Y_1 . Hence

$$\deg_U g_r \leq r - h_r, \quad \deg_{Y_1} g_r \leq M - h_r.$$

Multiplication by $(U - Y_1)^{h_r}$ does not mix monomials in $Y_2, \dots, Y_d$. Decomposing coefficientwise in those monomials therefore shows that every monomial of g_r has ω -weight at most $W + r$. Thus

$$G_r := T^r E^{h_r} g_r$$

belongs to $\mathcal{K}_r$, where on the right E denotes the polynomial in Equation (37). Its coefficient of T^r is exactly f_r . Both F and G_r lie in $\ker \Gamma$, so $F - G_r$ has strictly larger T -adic order and remains in the kernel. Iteration terminates after at most m steps and expresses F as an element of $\sum_r \mathcal{K}_r$. The independence already proved makes this sum direct, establishing Equation (41).

Finally, subtracting the dimensions in Equation (46) from Equation (45) gives $\text{rank}(\Gamma) = R_d(m, M, W)$. The factorization Equation (38) gives the last assertion. $\square$

The equality in Equation (42) concerns the enlarged, point-independent map Γ . It does not assert that $\Phi_{\alpha,y}$ has the same rank: the first substitution may have proper image, and

$$\text{rank}(\Phi_{\alpha,y}) = \text{rank}\left(\Gamma|_{\text{im } \Psi_{\alpha,y}}\right) \leq \text{rank}(\Gamma).$$

The next proposition, new to this work, gives an exact block decomposition of this restricted map.

4.4 A block decomposition of the actual local map

For nonnegative integers $u, t_0, \dots, t_d$, define the integer multinomial coefficient

$$\text{Mult}(u; t_0, \dots, t_d) := \begin{cases} \frac{u!}{t_0! \cdots t_d!}, & t_0 + \cdots + t_d = u, \\ 0, & \text{otherwise,} \end{cases} \quad (49)$$

with value zero as well if any t_j is negative.

Proposition 4.6 (Exact block decomposition of the local rank). *For a source monomial*

$$X^x Y_0^u Y_1^b Y^c \in Q(D, A, d, m, M, W),$$

define its two grades by

$$\ell = u + b + |c|, \quad \nu = x - b - \sum_{j=2}^d j c_j. \quad (50)$$

For an output monomial $T^r Z^s Y_1^e Y^z$, where $z = (z_2, \dots, z_d) \in \mathbb{N}^{d-1}$, define

$$\ell = s + e + |z|, \quad \nu = r - (d+1)s - e - \sum_{j=2}^d j z_j. \quad (51)$$

For each $(\ell, \nu) \in \mathbb{N} \times \mathbb{Z}$, let $\mathbf{C}_{\ell,\nu}$ be the matrix whose columns are the source monomials of grade (ℓ, ν) , whose rows are the output monomials of that grade satisfying

$$0 \leq r < m, \quad r \geq (d+1)s, \quad (52)$$

and whose entry in the row $T^r Z^s Y_1^e Y^z$ and column $X^x Y_0^u Y_1^b Y^c$ is the reduction in $\mathbb{F}_q$ of

$$(-1)^{\sum_{j=2}^d (j+1)(z_j - c_j)} \text{Mult}(u; s, e - b, z_2 - c_2, \dots, z_d - c_d). \quad (53)$$

When $d = 1$, the final list of differences in Equation (53) is empty. Then the exact rank of the actual local constraint map is independent of (α, y) and satisfies

$$\text{rank}_{\mathbb{F}_q}(\Phi_{\alpha,y}) = \sum_{\ell,\nu} \text{rank}_{\mathbb{F}_q}(\mathbf{C}_{\ell,\nu}). \quad (54)$$

The sum is finite. Moreover, a source monomial belongs to $\mathbf{C}_{\ell,\nu}$ only if

$$D\ell + \nu < mA. \quad (55)$$

Proof. First observe that the rank is independent of the received pair. The translation

$$Q(X, Y_0, Y_1, \dots, Y_d) \mapsto Q(X + \alpha, Y_0 + y, Y_1, \dots, Y_d)$$

is an automorphism of $Q(D, A, d, m, M, W)$: expanding the two translated variables can only lower their exponents and hence their weighted degree. Composing the normalized local map at $(0, 0)$ with this automorphism gives the map at (α, y) .

At $(0, 0)$, the local substitution is

$$X = T, \quad Y_0 = \sum_{j=1}^d (-1)^{j+1} T^j Y_j + TE.$$

Rows with $i < s$ are identically zero because every occurrence of E in the local substitution comes with a factor of T . Delete these rows and relabel each remaining target coordinate by replacing E with $T^d Z$. This is injective on the remaining target basis because

$$T^i E^s Y_1^e Y^z \mapsto T^{i+ds} Z^s Y_1^e Y^z, \quad i + ds < m.$$

Its image consists exactly of the rows with $r \geq (d+1)s$ in Equation (52). Thus it preserves rank and changes the normalized substitution to

$$Y_0 = \sum_{j=1}^d (-1)^{j+1} T^j Y_j + T^{d+1} Z \pmod{T^m}. \quad (56)$$

Expand a source monomial. To obtain $T^r Z^s Y_1^e Y^z$, the u factors of Y_0 must contribute s copies of $T^{d+1} Z$, $e - b$ copies of TY_1 , and $z_j - c_j$ copies of $(-1)^{j+1} T^j Y_j$ for each $2 \leq j \leq d$. The coefficient is exactly Equation (53), and the resulting exponent of T is

$$r = x + (d+1)s + (e - b) + \sum_{j=2}^d j(z_j - c_j). \quad (57)$$

The choice counts also give

$$u = s + (e - b) + \sum_{j=2}^d (z_j - c_j).$$

These two identities say precisely that both grades in Equations (50) and (51) are preserved. Conversely, when the lower arguments of the multinomial coefficient are nonnegative, these identities describe every term in the expansion.

Distinct grades have disjoint source bases and disjoint target coordinates. The matrix of the local map is therefore block diagonal with blocks $C_{\ell, \nu}$, proving Equation (54). Finally, the source weighted degree is

$$x + Du + (D-1)b + \sum_{j=2}^d (D-j)c_j = D\ell + \nu,$$

which proves Equation (55) and also makes the collection of nonempty blocks finite. $\square$

The block decomposition is not confined to this interpolation space. For any monomial interpolation support, including the weighted support of the main paper (Theorem 6.8), one obtains the exact local matrix by retaining precisely the columns belonging to that support and deleting empty rows.

Because the block entries are integers, ranks computed over $\mathbb{Q}$ are characteristic-independent upper bounds for the ranks in Equation (54). A rank computed modulo one prime is a lower bound for the rational rank, not by itself an upper bound in other characteristics. The exact block sum may be strictly smaller than $R_d(m, M, W)$; the latter remains the convenient closed-form upper bound used below. In particular, the exact block sum in Equation (54) may replace $R_d(m, M, W)$ in any finite interpolation certificate over a specified field.

4.5 The finite certificate

We now state a closed-form sufficient interpolation test. Its left side is the exact dimension, and its right side is the exact rank of the enlarged map Γ , hence an explicit upper bound on the actual local rank. The exact block sum from Proposition 4.6 gives a sharper field-specific test whenever its block rank is smaller. All of these quantities are finite, so no limit or hidden constant enters the assertion.

Theorem 4.7 (Finite interpolation certificate). *Assume the parameters in Equation (20) satisfy*

$$\sum_{\substack{c \in \mathbb{N}^{d-1} \\ \omega(c) \leq W}} \sum_{b_1=0}^M N_D \left(mA - (D-1)b_1 - \sum_{j=2}^d (D-j)c_j \right) > n \sum_{r=0}^{m-1} \Lambda_d(W+r) ((r+1)(M+1) - (r-h_r+1)_+ (M-h_r+1)_+). \quad (58)$$

Then, for every choice of distinct evaluation points $\alpha_1, \dots, \alpha_n \in \mathbb{F}_q$ and every received word $y \in \mathbb{F}_q^n$, there is a nonzero $Q \in \mathcal{Q}(D, A, d, m, M, W)$ satisfying the local constraints at all received points. Every $P \in \mathbb{F}_q[X]_{\leq D}$ with agreement at least A then satisfies

$$Q(X, P(X), P^{[1]}(X), \dots, P^{[d]}(X)) \equiv 0. \quad (59)$$

The interpolant also satisfies the total Y -degree bound in Equation (27).

Proof. By Proposition 4.3, the left side of Equation (58) is exactly $\dim \mathcal{Q}$. By Lemma 4.5, each received point imposes rank at most the sum on the right after removing the factor n . Finally, Lemma 4.2 supplies the specialization-degree hypothesis of Proposition 3.4. That proposition gives the nonzero interpolant and Equation (59). $\square$

5 One-sided supports and ambient-dimension estimates

This section retains the dimension and reachable-coordinate estimates for an alternative one-sided interpolation support. They isolate the effect of ambient padding and weighted-simplex enlargement, and supply context for Section C. The sharper weighted-support certificate is proved in the main paper; its proof is not repeated here.

Fix a gap $\delta \in (0, 1)$, and define the baseline ambient rate

$$\rho_0 := \frac{\delta(1-\delta)}{2}.$$

For a code of dimension k and block length n , set

$$A := k + \lceil \delta n \rceil, \quad K := \max\{k, \lceil \rho_0 n \rceil\}, \quad D := K - 1. \quad (60)$$

Throughout this section we assume $A \leq n$.

5.1 Uniform ambient geometry

Write $r = k/n$ and

$$\rho(r) := \max\{r, \rho_0\}, \quad \eta(r) := r + \delta.$$

The factor $1/2$ in this baseline leaves extra room at the low-rate endpoint. It is not needed for the qualitative all-rate result. The weighted-support proof in [Section 6.1](#) of the main paper uses the larger baseline $\delta/2$, implemented by $\lfloor \delta n/2 \rfloor$, and a separate endpoint analysis.

Lemma 5.1 (Adaptive ambient bounds). *Suppose $A \leq n$. Then $0 \leq r \leq 1 - \delta$, $k \leq K < A$, and*

$$\rho(r) \geq \frac{\delta(1-\delta)}{2}, \quad \frac{\eta(r)}{\rho(r)} \geq \frac{1}{1-\delta}, \quad 1 - \frac{\rho(r)}{\eta(r)} \geq \delta. \quad (61)$$

Moreover,

$$\frac{A}{D} > \frac{\eta(r)}{\rho(r)}$$

whenever $D > 0$, and $D/n \geq \rho_0 - 1/n$.

Proof. The inequality $A \leq n$ and the definition of A imply $k + \delta n \leq n$, hence $r \leq 1 - \delta$. If $K = k$, then $K < A$ because $\lceil \delta n \rceil \geq 1$. If $K = \lceil \rho_0 n \rceil$, then $K \leq \lceil \delta n \rceil < A$, where the strict inequality uses $k \geq 1$.

The first inequality in (61) is immediate. If $r \geq \rho_0$, then

$$\frac{\eta(r)}{\rho(r)} = \frac{r + \delta}{r} \geq \frac{1}{1 - \delta},$$

because $r \leq 1 - \delta$. If $r \leq \rho_0$, then

$$\frac{\eta(r)}{\rho(r)} = \frac{2(r + \delta)}{\delta(1 - \delta)} \geq \frac{2}{1 - \delta} > \frac{1}{1 - \delta}.$$

The third inequality is equivalent to $\rho(r) \leq (1 - \delta)(r + \delta)$. For $r \geq \rho_0$, this reduces to $r \leq 1 - \delta$; for $r \leq \rho_0$, it follows from $\rho_0 = \delta(1 - \delta)/2 \leq (1 - \delta)(r + \delta)$.

Finally, $A/n \geq \eta(r)$, while $D/n < \rho(r)$: in the branch $K = k$, we have $D = k - 1 < rn$, and in the other branch $D = \lceil \rho_0 n \rceil - 1 < \rho_0 n$. The lower bound on D/n follows from $K \geq \lceil \rho_0 n \rceil$. $\square$

The additive padding $K = k + \lfloor \lambda \delta n \rfloor$ also gives uniform separation, but it unnecessarily increases the ambient degree when k is already linear in n . The adaptive choice (60) uses padding only near rate zero and makes the elementary uniform bounds particularly simple. More generally, every fixed baseline in $(0, \delta(1 - \delta)]$ works qualitatively. We retain the present half-baseline for this existence proof and for the one-sided optimization in [Section C](#); it is not the ambient choice in the weighted-support certificate.

5.2 Dimension and coordinate estimates

We now compare the source dimension with the number of reachable local coordinates. Unlike BCPZZ's large-kernel analysis, the proof counts the local output coordinates directly. This shorter comparison isolates the two asymptotic effects that matter: the formal remainder exponent has about m/d choices, while enlarging the weighted simplex costs only $d^{1/a}$.

Let

$$\gamma := \frac{1}{1 - \delta},$$

and choose fixed real numbers $1 < a < b < \gamma$. For an integer $d \geq 3$, put

$$m = d^3, \quad H = H_{d-1} := \sum_{j=1}^{d-1} \frac{1}{j}, \quad W = \left\lfloor \frac{adm}{H} \right\rfloor, \quad C = \lceil bm \rceil.$$

For $c = (c_2, \dots, c_d) \in \mathbb{Z}_{\geq 0}^{d-1}$, recall

$$|c| = \sum_{j=2}^d c_j, \quad \omega(c) = \sum_{j=2}^d (j-1)c_j,$$

and define

$$\mathcal{B} = \{c \in \mathbb{Z}_{\geq 0}^{d-1} : \omega(c) \leq W, |c| \leq C\}.$$

Let $\mathcal{Q}_{\text{coarse}}$ be the span of the monomials

$$X^x Y_0^{b_0} Y_1^{b_1} Y^c, \quad c \in \mathcal{B}, \quad 0 \leq b_1 \leq m, \quad x, b_0 \in \mathbb{N}, \quad x + D(b_0 + b_1 + |c|) < mA. \quad (62)$$

Every specialization at a polynomial of degree at most D has degree less than mA .

The next dimension calculation reformulates the counting argument in [BCP⁺26, Lemma 3.2], with the ambient degree substituted for the actual message degree and the strict inequalities and rounding retained. It is governed by the function

$$f(u) := \frac{1}{2} \int_0^1 (u-z)_+^2 dz = \begin{cases} 0, & u \leq 0, \\ u^3/6, & 0 \leq u \leq 1, \\ (3u^2 - 3u + 1)/6, & u \geq 1. \end{cases} \quad (63)$$

Lemma 5.2 (Coarse dimension lower bound). *Suppose $D > d$. For all sufficiently large d , uniformly over all n, k with $A \leq n$,*

$$\dim \mathcal{Q}_{\text{coarse}} \geq |\mathcal{B}| D m^3 f\left(\frac{A}{D} - \frac{C}{m}\right).$$

Consequently, for all sufficiently large n ,

$$\frac{\dim \mathcal{Q}_{\text{coarse}}}{n} \geq \frac{\rho_0}{2} |\mathcal{B}| m^3 f\left(\frac{\gamma - b}{2}\right). \quad (64)$$

Proof. For fixed c, b_1 , put $L = mA - D(|c| + b_1)$. The number of pairs $(x, b_0) \in \mathbb{Z}_{\geq 0}^2$ with $x + Db_0 < L$ is at least $L_+^2/(2D)$. Since $|c| \leq C$, summing over $0 \leq b_1 \leq m$ and comparing the decreasing sum with its integral gives

$$\dim \mathcal{Q}_{\text{coarse}} \geq |\mathcal{B}| \frac{D}{2} \sum_{b_1=0}^m (mA/D - C - b_1)_+^2 \geq |\mathcal{B}| D m^3 f(A/D - C/m).$$

By Lemma 5.1, $A/D > \gamma$. Since $C/m \rightarrow b$ as $d \rightarrow \infty$, the argument of f is at least $(\gamma - b)/2$ for large d . The same lemma gives $D/n \geq \rho_0/2$ for large n , proving (64). $\square$

Define

$$\Lambda(z) := \#\{c \in \mathbb{Z}_{\geq 0}^{d-1} : \omega(c) \leq z\}.$$

Lemma 5.3 (Weighted-simplex enlargement). *For fixed $1 < a < b$ and $m = d^3$,*

$$\frac{\Lambda(W + m)}{|\mathcal{B}|} = O_{a,b}(d^{1/a}). \quad (65)$$

Proof. Consider the weighted simplex

$$S_W = \left\{ x \in \mathbb{R}_{\geq 0}^{d-1} : \sum_{j=1}^{d-1} jx_j \leq W \right\}.$$

Its volume is $W^{d-1}/((d-1)!)^2$, and a uniform point $X \in S_W$ satisfies $\mathbb{E}[|X|] = WH/d \leq am$. Markov's inequality shows that the portion with $|X| \leq C$ has volume at least $(1 - a/b) \text{vol}(S_W)$. Taking coordinatewise floors maps this portion into $\mathcal{B}$, so

$$|\mathcal{B}| \geq (1 - a/b) \frac{W^{d-1}}{((d-1)!)^2}.$$

Conversely, the disjoint unit cubes based at the lattice points counted by $\Lambda(W + m)$ lie in the simplex of radius $W + m + \binom{d}{2}$. Hence

$$\Lambda(W + m) \leq \frac{(W + m + \binom{d}{2})^{d-1}}{((d-1)!)^2}.$$

Since $W = (1 + o(1))adm/H$, their ratio is at most

$$\frac{b}{b-a} \left(1 + \frac{m + \binom{d}{2}}{W} \right)^{d-1} = O_{a,b}(e^{H/a}) = O_{a,b}(d^{1/a}).$$

□

Lemma 5.4 (Direct local-coordinate bound). *For every received pair (α, y) ,*

$$\text{rank}(\Phi_{\alpha,y}|_{Q_{\text{coarse}}}) \leq \left\lceil \frac{mA}{D} \right\rceil \sum_{r=0}^{m-1} \left\lceil \frac{m-r}{d+1} \right\rceil \Lambda(W+r). \quad (66)$$

If $m = d^3$, $d \geq 3$, and $n \geq 2/\rho_0$, then

$$\text{rank}(\Phi_{\alpha,y}|_{Q_{\text{coarse}}}) \leq \left(\frac{2}{\rho_0} + 1 \right) \frac{m^3}{d} \Lambda(W+m). \quad (67)$$

Consequently,

$$\frac{\text{rank}(\Phi_{\alpha,y}|_{Q_{\text{coarse}}})}{|\mathcal{B}|m^3} = O_{\delta,a,b} \left(d^{-(1-1/a)} \right). \quad (68)$$

Proof. Expand a source monomial after the local substitution and consider a retained output coordinate

$$T^i E^h Y_1^e Y^z.$$

Constants arising from α and y only lower source exponents. Every coordinate therefore satisfies

$$i \geq h, \quad \omega(z) \leq W + i - h, \quad i + dh < m, \quad h + e + |z| < \frac{mA}{D}. \quad (69)$$

Put $r = i - h$. The second inequality allows at most $\Lambda(W + r)$ choices for z . The local truncation becomes $r + (d+1)h < m$, so h has at most $\lceil (m-r)/(d+1) \rceil$ possible values. Finally, after fixing (r, h, z) , the last inequality allows at most $\lceil mA/D \rceil$ possible values for e . The rank is at most the number of reachable output coordinates, which proves (66).

For $m = d^3$ and $d \geq 3$,

$$\sum_{r=0}^{m-1} \left\lceil \frac{m-r}{d+1} \right\rceil \leq \frac{m(m+1)}{2d} + m \leq \frac{m^2}{d}.$$

Moreover, $D \geq \rho_0 n/2$ when $n \geq 2/\rho_0$, while $A \leq n$. Thus

$$\left\lceil \frac{mA}{D} \right\rceil \leq \left(\frac{2}{\rho_0} + 1 \right) m.$$

Monotonicity of Λ gives (67). Dividing by $|\mathcal{B}|m^3$ and applying Lemma 5.3 proves (68). $\square$

The one-sided weighted support and the comparison between a weighted simplex and its enlargement are adapted from BCPZZ. The ambient padding and the direct coordinate count are new.

Corollary 5.5 (Existence of a one-sided interpolation certificate). *For every fixed $\delta \in (0, 1)$, there exist integers*

$$d = d(\delta), \quad m = m(\delta), \quad W = W(\delta), \quad C = C(\delta),$$

and $N = N(\delta)$ such that, for every $n \geq N$, every $1 \leq k \leq n$ with $A = k + \lceil \delta n \rceil \leq n$, and the adaptive ambient dimension (60),

$$\dim \mathcal{Q}_{\text{coarse}} > n \operatorname{rank}(\Phi_{\alpha, y}|_{\mathcal{Q}_{\text{coarse}}})$$

for every received pair (α, y) . Hence a nonzero common interpolant exists, and every degree-at-most- D polynomial with at least A agreements satisfies its differential equation. The same derivative order $d(\delta)$ works for every k .

Proof. Choose $1 < a < b < 1/(1 - \delta)$. The constant on the right-hand side of (64) is positive. Since $1 - 1/a > 0$, the right-hand side of (68) tends to zero as $d \rightarrow \infty$. Choose one sufficiently large d , set $m = d^3$, and choose W, C as above. For all sufficiently large n , the lower bound for $\dim \mathcal{Q}_{\text{coarse}}/n$ in (64) is strictly larger than the local rank bound in (68). Stacking the n local maps therefore leaves a nonzero common kernel. The specialization-degree bound following (62), together with Proposition 3.4, proves the differential identity. All remaining requirements, including $D > d$ and the individual degree bounds needed by root finding, hold after increasing N , because $D/n \geq \rho_0 - 1/n$ and every interpolation parameter is fixed. $\square$

6 Small-order certificates and shrinking gaps

The main manuscript now gives first-order certificates at gaps down to 0.2195, together with linear list bounds and quadratic MCA bounds (Section 4.2). The certificates below retain simpler boundary cases and their original exact-rank calculations.

The following fixed-parameter certificates are not needed for the strongest derivative-order statement. They remain useful when one needs every interpolation parameter, including the multiplicity and jet degrees, to be independent of the block length.

Theorem 6.1 (Fixed-parameter small-order certificates). *The following interpolation certificates hold uniformly over the rate.*

1. Let $n \geq 2$, $A_0 = k + \lceil n/2 \rceil \leq n$, and $D_0 = k - 1$. The order-zero space

$$\mathcal{Q}_0 = \operatorname{span}\{X^u Y_0^b : b \in \{0, 1\}, u + D_0 b < A_0\} \quad (70)$$

has dimension $2A_0 - D_0 > n$, while each local map has rank one. The agreement list at threshold A_0 has at most one member.

2. Let $n \geq 512$, $A_1 = k + \lceil n/4 \rceil \leq n$, and

$$K_1 = \max\{k, \lfloor n/16 \rfloor\}, \quad D_1 = K_1 - 1, \quad \rho = D_1/n, \quad \gamma = A_1/D_1.$$

With derivative order $d = 1$, multiplicity $m = 64$, and Y_1 -degree bound $M = 16$, let

$$Q_1 = \text{span}\{X^u Y_0^a Y_1^b : 0 \leq b \leq 16, u + D_1 a + (D_1 - 1)b < 64A_1\}. \quad (71)$$

Its local rank is exactly 28152 in every characteristic, and

$$\frac{\dim Q_1}{n} \geq \rho \sum_{b=0}^{16} G(64\gamma - b) \geq \frac{585072}{19} > 28152. \quad (72)$$

Thus a nonzero order-one interpolant exists at every rate.

Proof. For the order-zero assertion,

$$\dim Q_0 = A_0 + (A_0 - D_0) = 2A_0 - D_0 > n.$$

The local map is evaluation at one received pair, so its rank is one because the space contains the constant polynomial. A nonzero interpolant has the form $Q_0(X) + Q_1(X)Y_0$. Here $Q_1 \neq 0$: otherwise the nonzero polynomial Q_0 , of degree less than $A_0 \leq n$, would vanish at all n distinct evaluation points. Any listed polynomial must therefore equal $-Q_0/Q_1$, proving uniqueness.

We turn to order one. First, $\gamma \geq 4/3$. If $K_1 = k$, this follows from $k \leq 3n/4$ and $A_1 \geq k + n/4$. If $K_1 > k$, then $D_1 < n/16$ and $A_1 \geq n/4$, so in fact $\gamma > 4$.

Translations in X and Y_0 act triangularly on (71), so they preserve the local rank. Normalize the received pair to $(0, 0)$. At T -degree $s \in \{0, \dots, 63\}$, the intermediate space is the rectangle

$$\text{span}\{T^s U^a Y_1^b : 0 \leq a \leq s, 0 \leq b \leq 16\}.$$

Every displayed monomial occurs in the image of the source: use $X^{s-a} Y_0^a Y_1^b$. Its source weight is at most

$$s + (D_1 - 1)(s + 16) \leq 79D_1 - 16 < 64A_1,$$

where the last inequality follows from $\gamma \geq 4/3$. The remaining substitution is $E = U - Y_1$. At T -degree s , its kernel on the rectangle consists exactly of the polynomials divisible by $(U - Y_1)^{64-s}$. Writing $h = 64 - s$, the total intermediate dimension is $17 \cdot 64 \cdot 65/2 = 35360$, while the kernel has dimension

$$\sum_{h=1}^{16} (65 - 2h)(17 - h) = 7208. \quad (73)$$

Thus the local rank is $35360 - 7208 = 28152$. This argument uses divisibility by a monic polynomial, so it is valid in every characteristic.

It remains to verify the uniform dimension lower bound. For an integer L , let $N_D(L)$ count $(u, a) \in \mathbb{N}^2$ with $u + Da < L$. Then

$$N_D(L) = DG(L/D).$$

Consequently,

$$\dim Q_1 = \sum_{b=0}^{16} N_{D_1}(64A_1 - (D_1 - 1)b) \geq D_1 \sum_{b=0}^{16} G(64\gamma - b),$$

which is the first inequality in (72).

Put $S(z) = \sum_{b=0}^{16} \mathbf{G}(z - b)$. For $z \geq 16$ and $j = \lfloor z \rfloor$, direct summation gives

$$S(z) = 17(j - 7)z - \frac{17}{2}j(j + 1) + 680. \quad (74)$$

If $K_1 = k$, then $\gamma \geq 1 + 1/(4\rho)$, so it suffices to minimize $F(\rho) = \rho S(64 + 16/\rho)$. On an interval where $j = \lfloor 64 + 16/\rho \rfloor$ is constant, its derivative has the sign of

$$-j^2 + 127j - 816.$$

The sign is positive for $85 \leq j \leq 120$ and negative for $j \geq 121$. Hence the minimum is at $64 + 16/\rho = 121$, or $\rho = 16/57$, where

$$\rho S(64 + 16/\rho) = \frac{16}{57} S(121) = \frac{585072}{19}.$$

If $K_1 > k$, then $\rho < 1/16$ and $\gamma \geq 1/(4\rho)$. Formula (74) shows that $\rho S(16/\rho)$ is decreasing on $(0, 1/16]$, and therefore

$$\rho S(16/\rho) \geq \frac{1}{16} S(256) = \frac{65637}{2} > \frac{585072}{19}.$$

This proves (72) and the theorem. $\square$

Corollary 6.2 (A logarithmically vanishing gap). *Fix $C > 3$, let $\delta_n = C/\log n$, and suppose $n \leq q = \Theta(n)$ is prime. For all sufficiently large n , combining the main paper's weighted-support certificate with the all-root enumeration procedure in Section 7.4 gives decoding uniformly over the rate at gap δ_n , with*

$$d = n^{1.5/C+o(1)}, \quad m = n^{3/C+o(1)} = o(n).$$

Its list size and running time are at most

$$\exp\left(n^{1.5/C+o(1)}\right).$$

Proof. Substitution into (87) of the main paper gives the displayed estimates for d and m . Write $b = B_{\text{jet}}(\delta_n)$. The parameter formulas give $m\delta_n^{-2} = n^{3/C+o(1)}$ and $b\delta_n^{-2} = n^{3/C+o(1)}$. Thus $C > 3$ makes both quantities $o(n)$, so the current finite hypothesis

$$n \geq \max\{\lceil m/\delta_n^2 \rceil, \lceil 4b/\delta_n^2 \rceil\}$$

holds eventually. The interpolation space has $q^{O(d)}$ monomials and constraints. The differential root-finding bound of Section 7.4, with extension degree two, gives list size $q^{2d+O(1)}$. Since $q = \Theta(n)$, both quantities are at most $\exp(n^{1.5/C+o(1)})$; polynomial-time linear algebra in their dense dimensions has the same asymptotic form. $\square$

7 Finite quantitative refinements

This section preserves finite constants and comparisons that supplement the main paper's asymptotic bounds. References prefixed by “main paper” point to the corresponding statements in that manuscript.

7.1 Concrete first-order constants

The exact support test of main-paper [Proposition 5.10](#) gives much smaller parameters than the uniform margin estimate. [Table 1](#) uses $n = 65,536$, $k = \rho n$, $\eta_1 = 0.01$, and $A = \lceil n(a_1(\rho) + \eta_1) \rceil$. The derivative-weighted support and its row-height test certify every displayed tuple; main-paper [Lemma 5.6](#) gives the list and exceptional-count bounds. Here $M < D = k - 1$, so the positive characteristic requirement is just $p > D$. The bounds hold for every prescribed evaluation set and received line over such a field.

ρ	A	(m, M, B, H)	list upper bound	$ \mathcal{E} $ upper bound
1/16	14,950	(20, 12, 72, 736)	71,457,911	$7.576 \cdot 10^{15}$
1/4	31,379	(28, 12, 53, 671)	117,645,549	$2.583 \cdot 10^{16}$
1/2	45,869	(24, 6, 33, 399)	55,807,550	$1.097 \cdot 10^{16}$

Table 1: First-order constants at gap $\eta_1 = 0.01$. All displayed bounds are rounded upward; the MCA error is at most $|\mathcal{E}|/q$. These agreements lie below Johnson, so this table addresses a different threshold from [Table 2](#).

The search used $m \leq 64$, integers M within four of $\lfloor \lambda_{\partial} m \rfloor$, and integers B within ten of $\lceil mA/D \rceil$. It selected these supports using the coarse factorwise envelope within that finite range. The table reevaluates them with main-paper [\(55\)](#), minimizing its threshold L exactly; it does not assert global optimality. The ordinary-tail alternative can have smaller exact constants even though the factorwise estimate has better asymptotic gap dependence.

7.2 Dominance over full differentiation

The main paper's ordinary-tail transfer retains the received-curve degree directly. The next result records its comparison with the coarser envelope obtained by differentiating through the ordinary tail at an ambient reconstruction degree.

Corollary 7.1 (Dominance over full differentiation). *Let $2 \leq k \leq K \leq n$ and retain the remaining hypotheses and notation of main-paper [Lemma 5.9](#). For every actual derivative degree $0 \leq d \leq B_{\partial}$ and every $k \leq L \leq A$, the curve bound $E_d^{(\ell)}(L)$, reconstructed through the actual message degree $D = k - 1$, is no larger than the following full-differentiation envelope. Let J'_1 and F'_1 be the degree budgets $D_{\text{joint}}(B_{\partial})$ and $D_{\text{fiber}}(B_{\partial})$ in that lemma, evaluated at $D' = K - 1$. With $\tau' = 2D' - 1$, $b_0 = B_{\text{jet}} - B_{\partial}$, and*

$$J'_0 = \ell \frac{b_0(b_0 + 1)}{2} + B_Z(b_0 + \tau' b_0^2), \quad F'_0 = \frac{b_0(b_0 + 1)}{2},$$

define

$$E_{\text{full}}^{(K, \ell)}(L) = B_Z + \frac{n - L + 1}{A - L + 1} \left(\frac{n - k + 1}{A - k + 1} J'_1 + J'_0 \right) + \ell(n - L) \left(\frac{n - k + 1}{L - k + 1} F'_1 + F'_0 \right).$$

Then $E_d^{(\ell)}(L) \leq E_{\text{full}}^{(K, \ell)}(L)$ for every d, L , and consequently

$$\max_{0 \leq d \leq B_{\partial}} \min_{k \leq L \leq A} E_d^{(\ell)}(L) \leq \min_{k \leq L \leq A} E_{\text{full}}^{(K, \ell)}(L), \quad (75)$$

so every certificate obtained from that envelope remains valid.

Proof. First compare the terminal tails at $D = k - 1$ and put $\tau = 2D - 1$. Full differentiation through an ordinary tail of degree b has fiber and joint degrees

$$F_{\text{old}} = \frac{b(b + 1)}{2}, \quad J_{\text{old}} = \ell \frac{b(b + 1)}{2} + B_Z(b + \tau b^2).$$

The ordinary transfer instead has $F_{\text{new}} = b$ and $J_{\text{new}} = \ell b + B_Z \Psi_D(b)$. For $b \geq 1$,

$$b + \tau b^2 - \Psi_D(b) - 2(b-1) = (b-1)(\tau(b-1) - 1) - 2(b-2D-1)_+ \geq 0.$$

Thus the joint-degree saving, whose incidence multiplier is at least one, pays the extra $2(b-1)B_Z$ preliminary exclusions. The fiber term also decreases. Minimizing the new tail over its independent threshold can only improve it. For $b = 0$, both tails cost B_Z .

Add the actual order-one stages. Promoting a missing order-one stage only increases the bound because $D_{\text{fiber}}(j, r; b, c) \geq j$ and $2bc - c^2 \geq b$, and all order-one incidence factors are at least their order-zero counterparts. Finally increase the reconstruction degree from $D = k-1$ to $K-1$. Both arguments in the definition of c , as well as b and $\ell + (2D-1)B_Z$, are nondecreasing in D ; so are $rb + (j-r)c$ and $2bc - c^2$. This proves the pointwise comparison for every d, L . Taking first the minimum over L and then the maximum over the possible actual degree gives the displayed inequality. If $k = K = n$, the full-code endpoint was handled directly in main-paper [Lemma 5.9](#). $\square$

7.3 Johnson constants and integer tuning

The two list estimates in main-paper (67) are classical. Writing $x = \sqrt{\rho_-}$, the pairwise-agreement Johnson argument gives

$$|\mathcal{L}_D(y, A)| \leq \frac{1-x}{2\eta_0} + \frac{1+x}{4x}.$$

Its leading constant is $1-x$ times that of the main paper's closed interpolation recipe.

Retain a, ρ_-, t , and B_{cf} from the closed bound in main-paper (68). Let $m_B = \max\{\lceil \sqrt{\rho_-}/\eta_0 \rceil, 3\}$ and $t_B = m_B + 1/2$. BCHKS give the following bound [[BCH⁺26](#), full version, Theorem 4.6]:

$$B_{\text{BCHKS}} = \frac{2t_B^5 + 3t_B(1-a)\rho_-}{3\rho_-^{3/2}}n + \frac{t_B}{\sqrt{\rho_-}}.$$

Since $t \leq t_B$, $t_B \geq 7/2$, and $0 < \rho_- < 1$, comparison with the positive leading term gives

$$\frac{B_{\text{cf}}}{B_{\text{BCHKS}}} < \frac{(8/3)nt^3/\rho_-}{(2/3)nt_B^5/\rho_-^{3/2}} \leq \frac{4\sqrt{\rho_-}}{t_B^2} < \frac{16}{49}.$$

This compares the printed real bounds before truncation by the field size; either probability bound can be vacuous when q is too small.

The closed recipe in the main paper need not minimize the exceptional count. Retaining the Guruswami–Sudan constraints and BCHKS weighted challenge count [[GS99](#), [BCH⁺26](#)], we instead choose integer cutoffs directly.

Proposition 7.2 (Exact ordinary interpolation certificate). *Under the code and agreement hypotheses of main-paper [Theorem 5.12](#), choose integers $m \geq 1$ and $1 \leq B \leq \lfloor (mA-1)/D \rfloor$. Put $u = \min\{B, m-1\}$ and*

$$N = \sum_{j=0}^B (mA - Dj), \quad W = \sum_{j=0}^B j(mA - Dj), \quad R = \sum_{b=0}^u (m-b), \quad T = \sum_{b=0}^u b(m-b).$$

If $N > nR$, set

$$H = \max \left\{ B, \left\lfloor \frac{W - nT}{N - nR} \right\rfloor \right\}. \quad (76)$$

There is a symbolic explainer with jet degree at most B , challenge degree at most H , and weighted specialization degree less than mA . It is nonzero at every challenge and sound after every field extension. Consequently the list bound is at most B , and main-paper (40) applies with $(B_{\text{jet}}, B_Z) = (B, H)$.

Proof. Give the coefficient of each base monomial $X^i Y^j$, with $i + Dj < mA$ and $j \leq B$, challenge degree at most $H - j$. A multiplicity output row of Y -degree b and X -degree less than $m - b$ has entries of challenge degree at most $j - b$, with zero entries when $j < b$. Apply main-paper [Lemma 3.12](#) with column cost j and row cost b . Since $H \geq B$, its unknown and equation counts are $(H + 1)N - W$ and $n((H + 1)R - T)$. The chosen height gives the strict surplus

$$(H + 1)(N - nR) > W - nT.$$

The lemma supplies a primitive kernel vector, hence an interpolant nonzero at every challenge. The ordinary multiplicity argument of main-paper [Proposition B.1](#) proves the explaining identity. $\square$

To identify the leading constant, write $x = \sqrt{D/n}$, $\widehat{a} = A/n$, and $\widehat{\eta} = \widehat{a} - x$. For the full cutoff $B = \lfloor (mA - 1)/D \rfloor$, the exact sums give

$$\begin{aligned} (N - nR)/n &= \frac{1}{2}m^2(\widehat{a}^2/x^2 - 1) - \frac{1}{2}m(1 - \widehat{a}) + O_x(1), \\ (W - nT)/n &= \frac{1}{6}m^3(\widehat{a}^3/x^4 - 1) + O_x(m). \end{aligned}$$

At fixed reduced rate and small gap, choose $m \sim cx(1 - x)/(2\widehat{\eta})$ for fixed $c > 1$. Then the surplus is positive and [\(76\)](#) gives

$$B \sim \frac{c(1 - x)}{2\widehat{\eta}}, \quad H \sim \frac{c^2 x(1 - x)^2}{12(c - 1)\widehat{\eta}^2}.$$

The leading transfer term is proportional to BH ; minimizing $c^3/(c - 1)$ at $c = 3/2$ gives

$$BH \sim \frac{9x(1 - x)^3}{32\widehat{\eta}^3}, \quad \frac{BH}{1/(24\widehat{\eta}^3)} = \frac{27}{4}x(1 - x)^3 + o(1) \leq \frac{729}{1024} + o(1).$$

The denominator is the closed recipe's leading product. Thus tuning improves the constant at every fixed rate, while the inverse-gap exponent remains three; finite claims use the integer sums above.

[Table 2](#) gives exact examples for $n = 65,536$, $\eta_0 = 0.01$, and $A = \lceil n(\sqrt{D/n} + 0.01) \rceil$. The bounded search checks $1 \leq m \leq 4m_0$ and every permitted B , where m_0 is the closed multiplicity; it makes no global optimality claim. The MCA columns use the unified transfer and are rounded down to valid integer bounds; the list column uses main-paper [\(67\)](#).

k/n	A	selected (m, B, H)	list	closed $ \mathcal{E} $	tuned $ \mathcal{E} $
1/16	17,038	(14, 57, 568)	38	4,046,431,259	2,498,629,121
1/4	33,423	(18, 36, 504)	25	8,278,798,980	3,383,852,708
1/2	46,996	(15, 21, 234)	15	13,537,540,252	1,448,631,664

Table 2: Finite Johnson constants. Divide exceptional counts by q for the MCA error probability.

7.4 Regular lifting and the separant chain

We count the solutions of a differential equation that agree with the received word, using Kopparty's reconstruction from a regular initial Taylor jet [[Kop15](#), Theorem 4.3, Theorem 4.4, and Corollary 4.5]. Below the characteristic, the required binomial coefficients are nonzero, so such a jet has at most one lift. An explicit derivative chain controls singular solutions; coefficient-space incidence then gives a list bound independent of the field size.

Recall the Hasse-derivative convention of [Definition 2.1](#) and the weighted degree of [Definition 2.4](#). For nonzero $Q \in \mathbb{F}_q[X, Y_0, \dots, Y_r]$, write

$$\deg_Y Q := \deg_{Y_0, \dots, Y_r} Q, \quad \ell_D(Q) := \text{wdeg}_{1, D, D-1, \dots, D-r} Q.$$

By (9), for every $P \in \mathbb{F}_q[X]_{\leq D}$,

$$\deg_X Q(X, P, P^{[1]}, \dots, P^{[r]}) \leq \ell_D(Q). \quad (77)$$

7.5 Unique lifting below the characteristic

We first show why regular initial Taylor data determine at most one solution. Below the characteristic, each new coefficient enters the recurrence with a nonzero multiplier, so we can recover it uniquely.

Lemma 7.3 (Unique lift from a regular jet). *Let E be a field of characteristic zero or $p > D$, let $0 \leq s \leq D$, and let*

$$R \in E[X, Y_0, \dots, Y_s].$$

Fix $\alpha, \beta_0, \dots, \beta_s \in E$ such that

$$R(\alpha, \beta_0, \dots, \beta_s) = 0 \quad \text{and} \quad \frac{\partial R}{\partial Y_s}(\alpha, \beta_0, \dots, \beta_s) \neq 0. \quad (78)$$

There is at most one $P \in E[X]_{\leq D}$ satisfying

$$R(X, P, P^{[1]}, \dots, P^{[s]}) \equiv 0 \quad (79)$$

and $P^{[j]}(\alpha) = \beta_j$ for every $0 \leq j \leq s$. Moreover, this polynomial can be reconstructed, or its nonexistence detected, using a number of field operations polynomial in D , $\ell_D(R)$, and the sparse support size and binary exponent lengths of R . Here coefficients in the arbitrary field E are accessed through field operations and equality tests; no bit encoding of E is assumed.

Proof. Write $P(\alpha + T) = \sum_{i=0}^D p_i T^i$, with $p_i = \beta_i$ for $i \leq s$. For $s < i \leq D$, changing p_i by γ changes the j -th Hasse derivative by

$$\gamma \binom{i}{j} T^{i-j}. \quad (80)$$

The first affected coefficient in the differential equation is at T^{i-s} , and its change is exactly

$$\gamma \binom{i}{s} \frac{\partial R}{\partial Y_s}(\alpha, \beta_0, \dots, \beta_s). \quad (81)$$

Terms with two changed factors have order at least $2(i-s) > i-s$; a lower derivative or a coefficient p_j with $j > i$ affects only higher orders. The displayed multiplier is nonzero by regularity and the characteristic hypothesis. Solving this linear equation successively therefore fixes every remaining coefficient. Truncated polynomial arithmetic and a final check of (79) give the asserted reconstruction or rejection and its polynomial cost. $\square$

7.6 Counting and enumerating solutions

A solution can be singular for the original equation yet regular for a later derivative. We use one deterministic chain to find that stage. Fix a nonzero $Q \in F[X, Y_0, \dots, Y_r]$ whose individual jet degrees are smaller than the positive characteristic, if any. Starting from $Q_0 = Q$, form a deterministic derivative chain. If Q_i depends on a jet variable, let

$$s_i := \max\{j : Q_i \text{ depends on } Y_j\}, \quad Q_{i+1} := \frac{\partial Q_i}{\partial Y_{s_i}}. \quad (82)$$

Every member is nonzero by the degree hypothesis, and each derivative lowers total jet degree. The chain therefore ends at a nonzero $Q_t \in F[X]$.

Every solution has a first nonzero specialization along this chain; the preceding equation vanishes and its separant does not. Under the hypotheses of [Lemma 7.3](#), a regular initial jet determines at most one solution. Enumerating all jets counts all differential roots; restricting them by received-word agreement gives the field-independent lists needed for decoding. For the algorithmic statements below, $\text{size}(Q)$ denotes the bit length of the sparse coefficient-list representation of Q , with exponents written in binary.

Fix a finite base field $\mathbb{F}_q$, a differential polynomial Q , and an ambient degree D , and write

$$\mathcal{R}_D(Q) := \{P \in \mathbb{F}_q[X]_{\leq D} : Q(X, P, P^{[1]}, \dots, P^{[r]}) \equiv 0\}. \quad (83)$$

Define its instance-sensitive singularity degree by

$$H_{\text{ch}}(Q) := \max_{1 \leq i \leq t} \ell_D(Q_i), \quad (84)$$

with value zero when $t = 0$. The ambient degree D is suppressed in the notation H_{ch} . Differentiating once lowers weighted degree by at least $D - r$, so

$$H_{\text{ch}}(Q) \leq \max\{0, \ell_D(Q) - (D - r)\}. \quad (85)$$

More precisely, if $s = \max\{j : Q \text{ depends on } Y_j\}$, then all later members of the chain have no larger weighted degree, and therefore

$$H_{\text{ch}}(Q) = \ell_D\left(\frac{\partial Q}{\partial Y_s}\right). \quad (86)$$

Theorem 7.4 (Instance-sensitive refinement of Kopparty's root bound). *Let $E_0 = \mathbb{F}_q$ have characteristic p , let $0 \leq r \leq D < p$, and let*

$$Q \in E_0[X, Y_0, \dots, Y_r] \setminus \{0\}$$

have individual Y_j -degrees smaller than p , and put $\Delta = \deg_Y Q$. Let E/E_0 be an extension of size $S = q^e$, and assume $S > H_{\text{ch}}(Q)$. Then

$$|\mathcal{R}_D(Q)| \leq \frac{S}{S - H_{\text{ch}}(Q)} \Delta S^r. \quad (87)$$

All solutions can be found deterministically in

$$(\Delta + 1)S^{r+2} \text{poly}(\text{size}(Q), D, \ell_D(Q), e, \log q)$$

bit operations.

Proof. If Q is independent of the jet variables, it is a nonzero polynomial in X and has no solution. Otherwise the chain terminates at a nonzero univariate polynomial. For a solution P , let $j(P) \geq 1$ be the first index such that

$$Q_{j(P)}(X, P, P^{[1]}, \dots, P^{[r]}) \neq 0.$$

The preceding specialization is identically zero. The nonzero one has degree at most $H_{\text{ch}}(Q)$, and is therefore nonzero at at least $S - H_{\text{ch}}(Q)$ values $\alpha \in E$. Call such an α a regular witness for P .

Map a regular pair (P, α) to

$$(\alpha, P(\alpha), P^{[1]}(\alpha), \dots, P^{[r]}(\alpha)) \in E^{r+2}.$$

This map is injective. Indeed, two solutions with different values of $j(P)$ contradict the first nonzero chain evaluation at their common jet. If the indices agree, put $F = Q_{j(P)-1}$ and $s = s_{j(P)-1}$. The two polynomials solve the same equation $F = 0$, have the same order- r jet, and satisfy

$$\frac{\partial F}{\partial Y_s}(\alpha, P(\alpha), \dots, P^{[s]}(\alpha)) \neq 0.$$

They coincide by [Lemma 7.3](#).

Fix $\alpha \in E$. If $Q(\alpha, Y_0, \dots, Y_r) = 0$ as a polynomial, specialization of X commutes with differentiation in every Y_j . Inductively, every $Q_i(\alpha, Y_0, \dots, Y_r)$ is therefore zero. No solution has a regular witness at this α . Otherwise, $Q(\alpha, Y_0, \dots, Y_r)$ is a nonzero polynomial of total degree at most Δ . Every witness jet is a zero of this polynomial. By the Schwartz–Zippel lemma [\[Sch80\]](#), applied with $h = r + 1$ variables and total-degree bound $b = \Delta$, there are at most ΔS^r such jets. Thus there are at most ΔS^{r+1} regular witnesses, while each solution has at least $S - H_{\text{ch}}(Q)$. Double counting proves (87).

For the algorithm, enumerate $\alpha \in E$ and every jet $(\beta_0, \dots, \beta_r) \in E^{r+1}$. Discard a pair unless $Q_0(\alpha, \beta) = 0$ and some later chain value is nonzero. Let $j \geq 1$ be its first nonzero chain index, set $F = Q_{j-1}$ and $s = s_{j-1}$, and use [Lemma 7.3](#) with $(\alpha, \beta_0, \dots, \beta_s)$ to reconstruct a degree-at-most- D solution of $F = 0$, if one exists. Check all prescribed order- r jet values, the original identity $Q = 0$, and membership of every coefficient in E_0 . Every true solution has a regular witness, so it is reconstructed at least once. Sort coefficient vectors and remove duplicates. There are S^{r+2} point-jet pairs, and every pair requires evaluating at most $\Delta + 1$ members of the derivative chain. Every remaining operation is polynomial in the displayed parameters. $\square$

Remark 7.5 (The exponent for a single equation). The equation $Q = Y_r$ has exactly q^r solutions of degree less than r , namely all polynomials of degree at most $r - 1$. Thus no bound for a generic single order- r differential equation can improve the base-field exponent below r .

Theorem 7.6 (Generic stratified differential root bound). *Let q be prime, let $0 \leq r \leq D < q$, and let*

$$Q \in \mathbb{F}_q[X, Y_0, \dots, Y_r] \setminus \{0\}.$$

Put

$$\Delta := \deg_Y Q, \quad \ell := \ell_D(Q), \quad h := \max\{0, \ell - (D - r)\}, \quad (88)$$

and assume $\Delta < q$. Let $e \geq 1$, set $S = q^e$, and suppose $S > h$. Then

$$|\mathcal{R}_D(Q)| \leq \frac{S}{S - h} \Delta S^r. \quad (89)$$

Proof. If $\Delta = 0$, then Q is a nonzero polynomial in X , so the root set is empty. Assume henceforth that $\Delta > 0$.

Set $E := \mathbb{F}_S$. We first bound all solutions in $E[X]_{\leq D}$; the desired $\mathbb{F}_q$ -solutions form a subset. Consider any nonzero iterated partial derivative R of Q , allowing $R = Q$. If R is independent of all Y -variables, then $R \in E[X] \setminus \{0\}$ and has no solutions. Otherwise, let s be the largest index such that R depends on Y_s . No assumption is made that R depends on the intermediate variables. Put

$$t := \deg_{Y_s} R, \quad \delta_R := \deg_Y R,$$

and define

$$R_a := \frac{\partial^a R}{\partial Y_s^a}, \quad 0 \leq a \leq t. \quad (90)$$

Since $t \leq \delta_R \leq \Delta < q$, the polynomial

$$R_t = t! \text{LC}_{Y_s}(R) \quad (91)$$

is nonzero and does not depend on Y_s .

Write $\mathcal{Z}_D(R)$ for the set of $P \in E[X]_{\leq D}$ satisfying

$$R(X, P, \dots, P^{[s]}) \equiv 0,$$

where variables absent from R are simply ignored. These solutions split into regular strata and a residual singular stratum. For $0 \leq a < t$, the a -th regular stratum consists of those $P \in \mathcal{Z}_D(R)$ for which

$$R_b(X, P, \dots, P^{[s]}) \equiv 0 \quad (0 \leq b \leq a), \quad R_{a+1}(X, P, \dots, P^{[s]}) \not\equiv 0. \quad (92)$$

The remaining solutions make every $R_1, \dots, R_t$ vanish identically and therefore belong to $\mathcal{Z}_D(R_t)$. Since R_t has active order strictly below s , or is independent of all Y -variables, this is the residual equation in the recursion.

Fix a solution P in a regular stratum and write

$$H_P(X) := R_{a+1}(X, P, \dots, P^{[s]}).$$

Every such separant is a nonzero positive-order partial derivative of the original Q . Differentiating with respect to Y_j lowers weighted degree by at least $D - j \geq D - r$. Together with (77), this gives

$$\deg H_P \leq \ell - (D - r) \leq h. \quad (93)$$

Thus $H_P(\alpha) \neq 0$ for at least $S - h$ values $\alpha \in E$.

We next count the regular solution-point pairs (P, α) , jointly across all regular strata. Fix $\alpha \in E$ and $\beta_0, \dots, \beta_{s-1} \in E$, and consider the univariate polynomial

$$f(Z) := R(\alpha, \beta_0, \dots, \beta_{s-1}, Z). \quad (94)$$

For a pair in the a -th stratum, with $z = P^{[s]}(\alpha)$, condition (92) says that

$$f(z) = f'(z) = \dots = f^{(a)}(z) = 0, \quad f^{(a+1)}(z) \neq 0.$$

Since $\deg f \leq t < q$, these are precisely the roots of f classified by their exact multiplicity. Across all $a < t$, there are at most t possible values of z . For each resulting regular initial jet, [Lemma 7.3](#), applied to the equation $R_a = 0$, gives at most one degree-at-most- D solution. Hence the number of regular solution-point pairs is at most

$$S \cdot tS^s. \quad (95)$$

On the other hand, every regular solution contributes at least $S - h$ pairs. The number of regular solutions is therefore at most

$$\frac{S}{S - h} tS^s. \quad (96)$$

It remains to account for the residual equation $R_t = 0$. Its total Y -degree satisfies

$$\deg_Y R_t \leq \delta_R - t. \quad (97)$$

The residual R_t is again an iterated partial derivative of Q , so the same separant-degree bound applies at every lower active order. If $s = 0$, then R_t has no Y -variables and contributes no residual solutions, so (96) already gives the desired bound. Suppose $s \geq 1$. Induction on the active order, with the base case of a nonzero polynomial in $E[X]$, now gives

$$|\mathcal{Z}_D(R)| \leq \frac{S}{S-h} \left(tS^s + (\delta_R - t)S^{s-1} \right) \quad (98)$$

$$\leq \frac{S}{S-h} \delta_R S^s. \quad (99)$$

If the active order of R_t is $s' < s$, the induction hypothesis gives the residual contribution $\frac{S}{S-h}(\delta_R - t)S^{s'}$, which is no larger than the term displayed above; if R_t has no Y -variables, its residual contribution is zero. Applying (99) to $R = Q$, whose active order is at most r , proves (89). $\square$

The preceding proof is constructive. We state the resulting algorithm separately because later sections use its running time as well as its list bound.

Corollary 7.7 (Constructive root enumeration). *Under the hypotheses of Theorem 7.6, one can enumerate $\mathcal{R}_D(Q)$ deterministically in*

$$(\Delta + 1)S^{r+2} \text{poly}(\text{size}(Q), D, \ell, e, \log q) \quad (100)$$

bit operations. In particular, if e, r, Δ are fixed and $\text{size}(Q), D, \ell \leq q^{O(1)}$, the running time is $q^{e(r+2)+O(1)}$.

Proof. If $\Delta = 0$, then Q is a nonzero polynomial in X , so the root set is empty and can be returned after reading Q . Assume $\Delta > 0$. Follow the residual chain (91). At each polynomial R and each regular stratum, enumerate $\alpha \in E$ and all initial jets in E^{s+1} , retain the regular jets, and apply the reconstruction in Lemma 7.3 to the equation $R_\alpha = 0$. Verify the original polynomial identity, retain only polynomials with coefficients in $\mathbb{F}_q$, and remove duplicates. The degrees t encountered along the residual chain sum to at most Δ , so there are at most Δ regular strata. Each stratum enumerates at most S^{r+2} point-jet pairs. Arithmetic and all identity checks take time polynomial in the displayed parameters. A representation of E can be constructed deterministically within the same bound, for example by enumerating monic degree- e polynomials and testing irreducibility. $\square$

7.7 Specialization to the interpolant

We finally record the form used by the decoder. It exposes the two field-size regimes without repeating the interpolation proof.

Corollary 7.8 (Root bound for the hidden-derivative interpolant). *Let $D > d$, and let*

$$Q \in \mathbb{F}_q[X, Y_0, \dots, Y_d] \setminus \{0\}$$

be an interpolant satisfying

$$L_Q := \ell_D(Q) \leq mA - 1. \quad (101)$$

Put

$$\Delta_Q := \deg_Y Q, \quad H_Q := \max\{0, L_Q - (D - d)\}. \quad (102)$$

Assume q is prime, $d \leq D < q$, and $\Delta_Q < q$. For every $e \geq 1$ with $q^e > H_Q$,

$$|\mathcal{R}_D(Q)| \leq \frac{q^e}{q^e - H_Q} \Delta_Q q^{ed}. \quad (103)$$

Moreover,

$$\Delta_Q \leq \left\lfloor \frac{mA - 1}{D - d} \right\rfloor. \quad (104)$$

If $D = K - 1$, then

$$H_Q \leq \max\{0, mA - K + d\}. \quad (105)$$

In particular, $q^e \geq 2H_Q$ implies

$$|\mathcal{R}_D(Q)| \leq 2\Delta_Q q^{ed}. \quad (106)$$

The solutions can be enumerated within the running time of [Corollary 7.7](#).

Proof. Every Y_j has weight at least $D - d$. Therefore the weighted-degree bound (101) implies (104). The remaining claims follow from [Theorem 7.4](#), using

$$L_Q - (D - d) \leq mA - 1 - (D - d) = mA - K + d$$

when $D = K - 1$. □

Remark 7.9 (Characteristic and source conventions). Kopparty’s Theorem 4.3 is stated with $\deg_{Y_j} Q < q$ [[Kop15](#)]. Its surrounding application is over a prime-order field. Read over a field of characteristic p , the recursive differentiation step requires $\deg_{Y_j} Q < p$, since it uses $t! \text{LC}_{Y_j}(Q) \neq 0$; the unique-lifting step likewise requires $D < p$. Our assumptions that q is prime and $D, \Delta < q$ make both requirements explicit. The restriction on D is substantive: over $\mathbb{F}_q$, the equation $P^{[1]} = 0$ has the q^2 solutions $P = aX^q + b$ of degree at most q . More generally, in characteristic p , every polynomial $P(X) = H(X^p)$ satisfies $P^{[1]} = 0$. Once $D \geq p$, the initial-jet uniqueness used above therefore fails. There is also a harmless typographical slip in the proof on p. 167: the nonvanishing displayed for Q should instead be assigned to the separant H , while $Q(\alpha, P(\alpha), \dots) = 0$ follows from the polynomial identity. The algorithm and its induction use these intended relations.

7.8 Primitive normalization

An optional preprocessing step can improve instance-specific parameters. View Q as a polynomial in the jet variables with coefficients in $\mathbb{F}_q[X]$, let $g(X)$ be the greatest common divisor of its nonzero coefficients, and put $Q^{\text{prim}} = Q/g$. Since $\mathbb{F}_q[X]$ is an integral domain,

$$\mathcal{R}_D(Q) = \mathcal{R}_D(Q^{\text{prim}}). \quad (107)$$

Passing to the primitive part cannot increase the jet degrees or weighted degree. The theorem below applies directly to either polynomial. If this optional normalization is used, its cost must be charged to the original input: grouping M sparse monomials by their jet exponents, computing coefficient gcds, and performing divisions takes $\text{poly}(M, \ell_D(Q), \log q)$ bit operations. The resulting coefficient lists contain at most $M(\ell_D(Q) + 1)$ monomials. Our decoder does not require this preprocessing.

8 Field descent and further refinements

The interpolation constraints depend only on the evaluation points and the received values. This lets us replace the ambient field by the subfield generated by the instance. We then return to Kopparty’s coefficient-lifting argument. Kopparty already branches at resonant Taylor indices in his general solver [[Kop15](#), Corollary 4.5]. We use that mechanism to give a refined root count for the present interpolants when the ambient degree is not below the characteristic.

8.1 Descent to the data field

Let p be the characteristic of $\mathbb{F}_q$. For an instance (α, y) , define its *data field* by

$$F_0 := \mathbb{F}_p(\alpha_1, \dots, \alpha_n, y_1, \dots, y_n) \subseteq \mathbb{F}_q, \quad q_0 := |F_0|. \quad (108)$$

Theorem 8.1 (Data-field descent). *Let $\alpha_1, \dots, \alpha_n \in \mathbb{F}_q$ be distinct, let $y \in \mathbb{F}_q^n$, and let $1 \leq k \leq A \leq n$. Set*

$$\mathcal{L}_{<k}(y, A) := \{P \in \mathbb{F}_q[X] : \deg P < k, \text{agr}((P(\alpha_i))_{i=1}^n, y) \geq A\}.$$

Then the following statements hold.

- (i) *Every polynomial in $\mathcal{L}_{<k}(y, A)$ belongs to $F_0[X]$, and $q_0 \geq n$.*
- (ii) *Every local interpolation system in [Sections 3 and 4](#) has coefficients in F_0 . Whenever the dimension comparison gives a nonzero interpolant over $\mathbb{F}_q$, it also gives one over F_0 . Root finding may therefore be performed over extensions of F_0 , and its field-size dependence may be expressed in terms of q_0 rather than q .*

Given a standard explicit representation of $\mathbb{F}_q$, a basis and arithmetic for F_0 can be constructed using a number of operations polynomial in n and $\log q$.

Proof. Fix $P \in \mathcal{L}_{<k}(y, A)$. Choose any k agreement positions. Lagrange interpolation at these positions expresses P using only the corresponding α_i 's, the values y_i , and inverses of their nonzero pairwise differences. Every such quantity lies in F_0 , which proves the descent in (i). The field F_0 contains the n distinct evaluation points, so $q_0 \geq n$.

For (ii), inspect the substitution in [Definition 3.2](#). All matrix entries are polynomial expressions in the α_i 's and y_i 's with coefficients in the prime field, and hence lie in F_0 . The rank of a matrix does not change after extending its coefficient field: its rank is the largest size of a nonzero minor. Thus a homogeneous system over F_0 that has a nonzero kernel vector over $\mathbb{F}_q$ already has a nonzero kernel vector over F_0 . The multiplicity argument is an identity over the same field. By (i), every requested message is also defined over F_0 , so the subsequent differential root search may take F_0 as its base field.

It remains to justify the algorithmic assertion. Write $s = [\mathbb{F}_q : \mathbb{F}_p]$, and start with the $\mathbb{F}_p$ -span of 1 and all input elements. Repeatedly enlarge the span whenever multiplying a current basis element by an input generator leaves the span. Each enlargement raises the dimension, so at most s enlargements occur. At termination the span is stable under all input generators, hence equals the $\mathbb{F}_p$ -algebra they generate. A finite subalgebra of a field is itself a field, so this algebra is F_0 : multiplication by any nonzero element is injective on the finite algebra and therefore surjective. Linear algebra over $\mathbb{F}_p$ supplies a basis, multiplication matrices, and inverses within the stated complexity. $\square$

8.2 Resonant coefficient lifting

The unique-lifting argument in [Lemma 7.3](#) solves for the Taylor coefficient p_i by dividing by $\binom{i}{s}$, where s is the active derivative order. When the characteristic does not exceed the ambient degree, the vanishing of this binomial coefficient does not destroy the recursion. It makes p_i a free branch, as in Kopparty's general coefficient-lifting procedure [[Kop15](#), Corollary 4.5]. Our new contribution is to convert these branches into the global root bounds below and specialize them to hidden-derivative interpolants. The following quantity counts those branches, including the s coefficients below the initial jet order:

$$\kappa_s(D, p) := |\{0 \leq i \leq D : \binom{i}{s} = 0 \text{ in characteristic } p\}|, \quad \kappa(D, p, d) := \max_{0 \leq s \leq d} \kappa_s(D, p). \quad (109)$$

Lemma 8.2 (Lifting with resonances). *Let E be a finite field of characteristic p and cardinality S . Let $0 \leq s \leq D$, and let $R \in E[X, Y_0, \dots, Y_s]$. Suppose that*

$$R(\alpha, \beta_0, \dots, \beta_s) = 0, \quad \frac{\partial R}{\partial Y_s}(\alpha, \beta_0, \dots, \beta_s) \neq 0.$$

Among the polynomials $P \in E[X]_{\leq D}$ with $P^{[j]}(\alpha) = \beta_j$ for $0 \leq j \leq s$, at most

$$S^{\kappa_s(D, p) - s} \quad (110)$$

satisfy $R(X, P, P^{[1]}, \dots, P^{[s]}) \equiv 0$. All of them can be found by successive coefficient lifting and branching only at the indices $i > s$ for which $\binom{i}{s} = 0$ in E .

Proof. Write $P(\alpha + T) = \sum_{i=0}^D p_i T^i$. The initial jet fixes $p_0, \dots, p_s$. Suppose the coefficients through p_{i-1} have been chosen. Changing p_i by c changes no coefficient below degree T^{i-s} , and the change in degree T^{i-s} is

$$c \binom{i}{s} \frac{\partial R}{\partial Y_s}(\alpha, \beta_0, \dots, \beta_s).$$

If $\binom{i}{s} \neq 0$, at most one value of c can make the coefficient vanish. If $\binom{i}{s} = 0$, retaining every value of $c \in E$ loses no solution. The binomial coefficient also vanishes for every $i < s$, while it equals one for $i = s$. The number of branching indices above s is therefore $\kappa_s(D, p) - s$. Continuing through $i = D$ and checking the full identity proves the claim. $\square$

We can now rerun the stratified root count without the hypothesis $D < p$. The remaining characteristic condition concerns only the individual degrees of the differential polynomial.

Theorem 8.3 (Differential root bound with resonances). *Let F_0 be a finite field of cardinality q_0 and characteristic p . Let $d \leq D$, and let*

$$Q \in F_0[X, Y_0, \dots, Y_d] \setminus \{0\}$$

have individual Y_j -degrees smaller than p . Put

$$\Delta := \deg_Y Q, \quad \ell := \ell_D(Q), \quad h := \max\{0, \ell - (D - d)\}, \quad \kappa := \kappa(D, p, d). \quad (111)$$

Let E/F_0 be an extension of cardinality $S = q_0^e$, and assume $S > h$. Then

$$\left| \{P \in F_0[X]_{\leq D} : Q(X, P, P^{[1]}, \dots, P^{[d]}) \equiv 0\} \right| \leq \frac{S}{S - h} \Delta S^\kappa. \quad (112)$$

All such polynomials can be enumerated deterministically in

$$(\Delta + 1) S^{\kappa+2} \text{poly}(\text{size}(Q), D, \ell, e, \log q_0) \quad (113)$$

bit operations.

Proof. We count all solutions over E , which can only enlarge the desired set. Consider any nonzero iterated partial derivative R of Q . If R contains no jet variable, then it is a nonzero polynomial in X and has no differential root. Otherwise, let s be its largest active jet order and set $t = \deg_{Y_s} R$. The hypothesis $t < p$ implies that $R_t := \partial^t R / \partial Y_s^t$ is nonzero and has active order below s .

As in the proof of [Theorem 7.6](#), the roots of R split into t regular multiplicity strata and the residual roots of R_t . For a root in a regular stratum, the corresponding separant is a nonzero positive-order partial derivative of Q . Its specialization has degree at most h , so the root has at least $S - h$ regular witnesses $\alpha \in E$.

Fix one such α and the lower jet $(\beta_0, \dots, \beta_{s-1})$. The possible values of β_s , counted jointly over the regular strata, are the roots of the nonzero univariate fiber

$$R(\alpha, \beta_0, \dots, \beta_{s-1}, Z),$$

classified by exact multiplicity. There are at most t of them. For each regular initial jet, [Lemma 8.2](#) leaves at most $S^{\kappa_s(D,p)-s}$ lifts. Hence the number of regular solution-witness pairs for R is at most

$$S \cdot S^s \cdot t \cdot S^{\kappa_s(D,p)-s} \leq tS^{\kappa+1}.$$

Double counting the witnesses bounds the regular roots by $\frac{S}{S-h}tS^\kappa$.

The total Y -degree of the residual polynomial R_t is at most $\deg_Y R - t$. Induction on the active order, starting with a nonzero polynomial in X , therefore gives

$$|\mathcal{Z}_D(R)| \leq \frac{S}{S-h}(t + (\deg_Y R - t))S^\kappa.$$

Taking $R = Q$ proves [\(112\)](#).

For enumeration, follow the same residual recursion. At each regular stratum, enumerate α , the lower jet, and β_s , and branch at precisely the resonant indices in [Lemma 8.2](#). The number of resulting trials per stratum is at most $S^{\kappa+2}$. The degrees encountered along the residual recursion sum to at most Δ . Checking the original identity, retaining the polynomials over F_0 , and removing duplicates gives [\(113\)](#). $\square$

The resonance count has a closed form in the range used by the decoder.

Proposition 8.4 (Closed form for the resonance count). *Assume $d < p$, and write $D = up + v$ with $0 \leq v < p$. Then, for $0 \leq s \leq d$,*

$$\kappa_s(D, p) = us + \min\{s, v + 1\}, \quad \kappa(D, p, d) = ud + \min\{d, v + 1\}. \quad (114)$$

In particular, $D < p$ gives $\kappa = d$, recovering the exponent in [Theorem 7.6](#).

Proof. For $i = ap + b$ with $0 \leq b < p$, the characteristic- p identity

$$(1 + Z)^i = (1 + Z^p)^a(1 + Z)^b$$

shows that $\binom{i}{s} = \binom{b}{s}$ for $s < p$. The latter coefficient is nonzero exactly when $b \geq s$. Each of the u complete residue blocks therefore contributes s zeros, and the last block contributes $\min\{s, v + 1\}$. This proves the first formula. Its right side is nondecreasing in s , so the maximum over $s \leq d$ occurs at d . If $D < p$, then $u = 0$ and $v = D \geq d$, giving $\kappa = d$. $\square$

The maximum resonance count and the single worst separant degree in [Theorem 8.3](#) are convenient but can be unnecessarily pessimistic. The residual chain yields the following term-by-term form.

Proposition 8.5 (Residual-chain root bound). *Under the hypotheses of [Theorem 8.3](#), define $R_0 = Q$. While R_j depends on a jet variable, let*

$$s_j = \max\{s : R_j \text{ depends on } Y_s\}, \quad t_j = \deg_{Y_{s_j}} R_j, \quad R_{j+1} = \frac{\partial^{t_j} R_j}{\partial Y_{s_j}^{t_j}},$$

and put

$$h_j = \ell_D \left(\frac{\partial R_j}{\partial Y_{s_j}} \right).$$

If $S > \max_j h_j$, then

$$|\mathcal{R}_D(Q)| \leq \sum_j \frac{S}{S-h_j} t_j S^{\kappa_{s_j}(D,p)}. \quad (115)$$

Proof. At stage j , the roots outside the residual equation $R_{j+1} = 0$ split among the t_j regular multiplicity strata. Every corresponding separant is a positive Y_{s_j} -derivative of R_j , so its specialization degree is at most h_j . The witness double count in the proof of [Theorem 8.3](#), now retaining the active order, therefore bounds these roots by

$$\frac{S}{S - h_j} t_j S^{\kappa_{s_j}(D, p)}.$$

The remaining roots satisfy $R_{j+1} = 0$. Iterating until the residual is a nonzero polynomial in X , which has no differential roots, and adding the regular contributions proves (115). $\square$

Proposition 8.6 (Exactness of the resonance exponent). *For $0 \leq s \leq D$, the equation $Y_s = 0$ over a field of cardinality q_0 and characteristic p has exactly*

$$q_0^{\kappa_s(D, p)} \quad (116)$$

polynomial solutions of degree at most D .

Proof. Writing $P = \sum_{i=0}^D a_i X^i$ gives

$$P^{[s]} = \sum_{i=s}^D a_i \binom{i}{s} X^{i-s}.$$

Distinct i 's give distinct monomials. Thus a_i is forced to vanish when $\binom{i}{s} \neq 0$ and is free when $\binom{i}{s} = 0$, including every $i < s$. The number of free coefficients is precisely $\kappa_s(D, p)$. $\square$

The closed form lets us decode when the characteristic grows only linearly with the block length, at a cost polynomial in the field size. We use the fixed-parameter certificates in this statement so that the individual jet degrees remain bounded independently of n .

Corollary 8.7 (Extension to linearly growing characteristic). *Fix $\delta \in (0, 1)$ and $c > 0$. There is $N_{\delta, c}$ and a deterministic algorithm such that, for every $n \geq N_{\delta, c}$, every finite field $\mathbb{F}_q$ of characteristic $p \geq cn$ and cardinality $q \geq n$, every integer $1 \leq k \leq n$, every set of n distinct evaluation points, and every received word, the algorithm outputs exactly all degree- $< k$ polynomials having at least $k + \lceil \delta n \rceil$ agreements. Its list size and running time are $q^{O_{\delta, c}(1)}$.*

More precisely, for the interpolant selected in the proof, of derivative order d and total jet degree Δ_Q , the differential-root stage has list bound

$$2\Delta_Q q^{2d\lceil 1/c \rceil} \quad (117)$$

for all sufficiently large n , with the convention that the exponent is zero when $d = 0$.

Proof. Put $A = k + \lceil \delta n \rceil$. If $A > n$, return the empty list. Otherwise the certified threshold used below is at most A and at most n . If $\delta \geq 1/2$, use the order-zero fixed-parameter certificate in [Theorem 6.1](#). If $1/4 \leq \delta < 1/2$, use its order-one certificate at threshold $k + \lceil n/4 \rceil$. If $0 < \delta < 1/4$, use the weighted-support certificate of the main paper ([Theorem 6.8](#)). In all three cases, d and every individual jet degree are bounded in terms of δ alone. Interpolation and multiplicity are characteristic-free. Once n is sufficiently large, p exceeds d and every individual jet degree, so the resonant root theorem applies.

Since $D < n$ and $p \geq cn$, writing $D = up + v$ gives

$$u < 1/c, \quad \kappa(D, p, d) \leq d(u + 1) \leq d\lceil 1/c \rceil$$

by [Proposition 8.4](#). The weighted specialization degree is $O_{\delta}(n)$, whereas $q^2 \geq n^2$; hence the extension $S = q^2$ satisfies $S \geq 2h$ for all sufficiently large n . Applying [Theorem 8.3](#) proves (117); its constructive part and the polynomial-size interpolation systems give the stated running time. Final degree and agreement filtering proves exactness. $\square$

8.3 Random regular witnesses

Deterministic enumeration spends one factor of S on the top entry of a regular jet. Randomly sampling the witness point and finding the roots of the resulting univariate fiber removes that factor. We state the result as a one-sided randomized algorithm: every output is checked, while completeness holds with prescribed probability.

Theorem 8.8 (Randomized regular-witness enumeration). *Adopt the hypotheses and notation of Theorem 8.3, and assume $S \geq 2h$. Given arithmetic in E , for every $\varepsilon \in (0, 1)$, there is a randomized algorithm that outputs only members of the differential root set in (112) and outputs every member with probability at least $1 - \varepsilon$. Its expected running time is*

$$S^\kappa \text{poly}(\text{size}(Q), D, \ell, \Delta, e, \log q_0, \log(1/\varepsilon)). \quad (118)$$

The same algorithm may check a degree restriction and an agreement threshold before output, without changing the displayed field-size exponent.

Proof. The root bound gives at most $M_0 := \max\{1, 2\Delta S^\kappa\}$ solutions. Every solution has at least $S - h \geq S/2$ regular witness points in the stratum where it leaves the residual chain. Sample

$$T := \lceil \log_2(M_0/\varepsilon) \rceil$$

independent uniform elements of E . A fixed solution has no sampled regular witness with probability at most 2^{-T} . A union bound over all solutions makes the probability of missing any solution at most ε .

It remains to explain how one sampled point is processed without enumerating the top jet entry. At each residual polynomial R of active order s , enumerate the S^s lower jets and form the fiber

$$f(Z) = R(\alpha, \beta_0, \dots, \beta_{s-1}, Z).$$

If $f = 0$, this lower jet gives no regular witness for R and may be skipped. Otherwise, find all roots of f in E , classify them by exact multiplicity, and apply the resonant lifting procedure. The roots can be found by a Las Vegas computation polynomial in $\deg f$, the coefficient-list size of f , and $\log S$: compute $\gcd(f, Z^S - Z)$ by modular exponentiation, then recursively split the resulting square-free product of linear factors. For a factor of degree r , evaluation of a uniformly random polynomial of degree below r at its r distinct roots gives independent uniform field values, by Vandermonde interpolation. In odd characteristic, random quadratic-character tests separate any two factors with constant probability. In characteristic two, the absolute trace to $\mathbb{F}_2$ gives the analogous split. Repeating only failed splits preserves correctness and gives polynomial expected time.

For a fiber of degree t , at most t top jet values survive. The lower-jet choices and the resonant branches number at most

$$S^s S^{\kappa_s(D,p)-s} \leq S^\kappa.$$

The degrees of the fibers along the residual chain sum to at most Δ . The number of samples is $T = O(\log \Delta + \kappa \log S + \log(1/\varepsilon))$, which is absorbed by the polynomial factor in (118). This proves the expected running-time bound. Every candidate is verified against the original identity, its base field, and any requested agreement condition, so the algorithm has no false outputs.

In the decoder, the extension degree is at most two. When $e = 2$, sample monic quadratics over F_0 and test irreducibility by a greatest common divisor with $Z^{q_0} - Z$. Exactly $(q_0^2 - q_0)/2$ of the q_0^2 monic quadratics are irreducible, as follows by pairing the elements of $\mathbb{F}_{q_0^2} \setminus F_0$ under Frobenius conjugation. Thus this constructs E in expected time polynomial in $\log q_0$ and does not change (118). $\square$

For an interpolant produced from the received instance, the multiplicity argument places the requested agreement list inside its differential root set. Combining [Theorems 8.1, 8.3 and 10.1](#) therefore gives three simultaneous list bounds. One may take the minimum of the differential bound (112), the subset bound (126), and, when its denominator is positive, the Johnson-type bound (127). Over a prime field with $D < q$, the data field is the full field and $\kappa = d$. Over a proper extension field, descent can reduce the base of the exponential root-finding term from q to q_0 .

9 Small characteristic and the Frobenius obstruction

The local interpolation and multiplicity arguments are identities over an arbitrary field. Small characteristic enters only when the differential roots are extracted. This section separates two failures of coefficient lifting from an obstruction for differential polynomials whose monomials all contain a higher derivative. The Frobenius decomposition used in the analysis is standard and is proved here for completeness. The obstruction identifies a support restriction that can prevent efficient enumeration of roots.

9.1 Two failures of coefficient lifting

There are two distinct reasons that the below-characteristic proof in [Section 7.4](#) cannot be applied verbatim.

Proposition 9.1 (Characteristic failures). *Over a field of characteristic p , the following statements hold.*

- (i) *Formal differentiation of a nonconstant jet polynomial can vanish. For example, $\partial(Y_1^p)/\partial Y_1 = 0$. Thus a nonzero recursive derivative chain is guaranteed by the hypothesis that every individual Y_j -degree is smaller than p .*
- (ii) *Even with a nonzero separant, an initial jet need not determine a unique polynomial once $D \geq p$. The equation $Y_1 = 0$ is satisfied by every $P(X) = H(X^p)$, since $P^{[1]} = 0$.*

Proof. The first assertion follows from $pY_1^{p-1} = 0$. For the second, the identity $(X + T)^p = X^p + T^p$ shows that the coefficient of T in $H((X + T)^p)$ is zero. Hence the first Hasse derivative vanishes, even though the separant of Y_1 is one. Equivalently, the coefficient lifting step at Taylor degree p encounters $\binom{p}{1} = 0$. $\square$

The first failure is excluded by the individual-degree hypothesis in [Theorem 8.3](#). The second is measured exactly by the resonance count $\kappa(D, p, d)$. In a fixed small characteristic, that count can grow linearly with D , so the resulting enumeration may still be exponential in the block length.

9.2 Frobenius coordinates

The behavior of Hasse derivatives in small characteristic is transparent after grouping monomials by their exponent modulo a power of the characteristic. We record the standard decomposition and its short proof to fix the degree conventions used in the obstruction.

Lemma 9.2 (Frobenius decomposition). *Let F have characteristic p , and let $M = p^u$. Every polynomial $P \in F[X]_{\leq D}$ has a unique representation*

$$P(X) = \sum_{a=0}^{M-1} X^a H_a(X^M), \quad \deg H_a \leq \left\lfloor \frac{D-a}{M} \right\rfloor \quad (119)$$

with the convention that $H_a = 0$ when $a > D$. For $0 \leq j < M$, its Hasse derivatives are

$$P^{[j]}(X) = \sum_{a=j}^{M-1} \binom{a}{j} X^{a-j} H_a(X^M). \quad (120)$$

Proof. Collect the monomials of P according to their exponent modulo M . This gives (119), its degree bounds, and its uniqueness. Since M is a power of p ,

$$(X + T)^M = X^M + T^M.$$

For $j < M$, the factor $H_a(X^M + T^M)$ contributes no positive power T^j . Taking the coefficient of T^j in $(X + T)^a H_a((X + T)^M)$ therefore gives $\binom{a}{j} X^{a-j} H_a(X^M)$. Summing over a proves (120). $\square$

The simultaneous kernel of the derivatives of orders $2, \dots, d$ has the same simple form in every characteristic.

Lemma 9.3 (The higher-derivative Frobenius kernel). *Let F have characteristic p , let $d \geq 2$, and let $M = p^u$ be the least power of p exceeding d . Then*

$$P^{[j]} = 0 \text{ for every } 2 \leq j \leq d \iff P(X) = H_0(X^M) + XH_1(X^M) \quad (121)$$

for uniquely determined $H_0, H_1 \in F[X]$.

Proof. The reverse implication follows at once from (120), since only the residue classes zero and one occur and $d < M$. For the forward implication, the minimality of M gives $p^{u-1} \leq d$. Hence $P^{[p^t]} = 0$ for every $1 \leq t < u$. If $a = \sum_{t=0}^{u-1} a_t p^t$, the characteristic- p factorization

$$(1 + Z)^a = \prod_{t=0}^{u-1} (1 + Z^{p^t})^{a_t}$$

gives $\binom{a}{p^t} = a_t$ in F : factors with lower index have combined degree at most $p^t - 1$, and factors with higher index have degree larger than p^t . In (120), distinct residue classes a contribute to distinct residue classes $a - p^t$, so no cancellation is possible. Thus every residue class in the support of P has $a_t = 0$ for $1 \leq t < u$. If $p = 2$, its remaining digit is already zero or one. If $p > 2$, the same factorization and the equation $P^{[2]} = 0$ give $\binom{a_0}{2} = 0$, so again $a_0 \in \{0, 1\}$. The Frobenius decomposition now gives the stated form and its uniqueness. $\square$

9.3 An obstruction for higher-derivative-supported equations

The preceding Frobenius identities force many roots of any equation whose monomials all contain a higher derivative. We examine the consequence for a decoder that enumerates all differential roots before filtering by agreement.

Call a differential polynomial *higher-derivative supported* if

$$Q \in (Y_2, \dots, Y_d) \subseteq F[X, Y_0, \dots, Y_d]. \quad (122)$$

Equivalently, every monomial of Q contains some Y_j with $j \geq 2$. The support in Theorem 6.8 of the main paper does not impose this restriction, so the obstruction below does not apply to every interpolant it supplies.

Theorem 9.4 (Frobenius-plane obstruction). *Let $F = \mathbb{F}_q$ have characteristic p , let $d \geq 2$, and let M be the least power of the characteristic exceeding d . Let*

$$Q \in F[X, Y_0, \dots, Y_d]$$

be higher-derivative supported. For every pair $H_0, H_1 \in F[X]$, the polynomial

$$P(X) = H_0(X^M) + XH_1(X^M)$$

satisfies

$$Q(X, P, P^{[1]}, \dots, P^{[d]}) \equiv 0. \quad (123)$$

Consequently, for every $D \geq 0$, the polynomial Q has at least

$$q^{\lfloor D/M \rfloor + \lfloor (D-1)/M \rfloor + 2} \quad (124)$$

differential roots of degree at most D .

Proof. By [Lemma 9.3](#), every derivative $P^{[j]}$ with $2 \leq j \leq d$ is zero. Each monomial of Q contains one of the corresponding variables, which proves (123). Under the degree restriction, one may choose H_0 and H_1 independently with

$$\deg H_0 \leq \left\lfloor \frac{D}{M} \right\rfloor, \quad \deg H_1 \leq \left\lfloor \frac{D-1}{M} \right\rfloor.$$

The decomposition is unique, so these choices give exactly the number of distinct polynomials displayed in (124), where $\lfloor -1/M \rfloor = -1$ when $D = 0$. $\square$

The next instance shows that this differential root set can be exponentially larger than the agreement list.

Corollary 9.5 (A singleton list with many differential roots). *Fix a prime p , fix $d \geq 2$, and let M be as in [Theorem 9.4](#). Let $q = n = p^s$, with n sufficiently large. Evaluate at every point of $\mathbb{F}_q$, take $k = \lfloor n/2 \rfloor$, $D = k - 1$, and take the all-zero received word. For every $\delta \in (0, 1/2]$, the agreement threshold $A = k + \lceil \delta n \rceil$ is at most n , and its list consists only of the zero polynomial. On the other hand, every nonzero higher-derivative-supported Q has at least*

$$q^{n/M} \quad (125)$$

differential roots of degree at most D .

Proof. A nonzero polynomial of degree less than k has at most $k - 1$ zeros, whereas $A > k - 1$. Thus only the zero polynomial reaches the agreement threshold. If $p = 2$, then $2M \mid n$ for all sufficiently large s , and the exponent in (124) is exactly n/M . If p is odd, write $n = ML$, where L is odd. Substituting $D = (n - 3)/2$ shows that the same exponent is L when $M = 3$ and $L + 1$ when $M \geq 5$. In every case it is at least n/M , proving (125). $\square$

Scope of the obstruction. [Corollary 9.5](#) rules out a specific algorithmic architecture over a full-field family in every fixed characteristic: construct a higher-derivative-supported interpolant, enumerate all of its bounded-degree differential roots, and only then filter by agreement. It does not give a large Reed–Solomon list. The exhibited instance has a singleton list. It also does not rule out a support that contains monomials outside the ideal $(Y_2, \dots, Y_d)$, a solver that uses the agreement constraints while extracting roots, or another capacity decoder over fixed-characteristic fields. The interpolation and multiplicity statements in [Section 3](#) remain valid in every characteristic.

10 Why a positive capacity gap is necessary

The decoder in [Theorem 9.16](#) of the main paper requires a strictly positive agreement gap above the message dimension. This requirement is not merely an artifact of the interpolation argument. We give a bad received word for every evaluation set. The same construction also quantifies what happens when the gap tends to zero with the block length.

Large lists and negative results for Reed–Solomon list decoding have a substantial literature; see, for example, [\[RR03, GR06, CW07, BKR10\]](#). We prove the exact formulas below directly and do not rely

on those earlier constructions. We make no priority claim for the elementary exact-capacity center. The coefficient-fiber formulation is included because it applies to every evaluation set and exposes the inverse-gap dependence needed here. The coefficient-pigeonhole construction uses no property of the evaluation set beyond distinctness; later examples use the full field or its multiplicative group.

10.1 Field-independent upper bounds

We first record two upper bounds that apply to every evaluation set and field. The subset count is elementary; the second is the standard Johnson-type incidence estimate. See [GRS25, Chapter 7] for background. We include the proofs to fix the finite conventions.

Theorem 10.1 (Combinatorial agreement bounds). *Let $1 \leq k \leq A \leq n$. Then*

$$|\mathcal{L}_{k-1}(y, A)| \leq \left\lfloor \frac{\binom{n}{k}}{\binom{A}{k}} \right\rfloor. \quad (126)$$

If $A^2 > n(k-1)$, then also

$$|\mathcal{L}_{k-1}(y, A)| \leq \left\lfloor \frac{n(A-k+1)}{A^2 - n(k-1)} \right\rfloor. \quad (127)$$

Proof. A k -subset of coordinates can lie in the agreement sets of at most one listed polynomial: two degree- $< k$ polynomials agreeing with y there would agree with each other at k points. Every list member contributes at least $\binom{A}{k}$ such subsets, while only $\binom{n}{k}$ exist. This proves (126).

For the second bound, put $L = |\mathcal{L}_{k-1}(y, A)|$, and choose exactly A agreement positions S_j for each list member. Let $m_i = |\{j : i \in S_j\}|$. Distinct chosen sets meet in at most $k-1$ positions, so

$$\sum_{i=1}^n m_i(m_i - 1) \leq L(L-1)(k-1). \quad (128)$$

Since $\sum_i m_i = LA$, Cauchy–Schwarz gives $\sum_i m_i^2 \geq L^2 A^2 / n$. Substitution in (128) yields

$$L \left(\frac{A^2}{n} - (k-1) \right) \leq A - k + 1.$$

The coefficient of L is positive by hypothesis, and rearranging proves (127). $\square$

These bounds constrain the agreement list rather than the differential-root set. They may be intersected with any algebraic root bound and remain valid when the present differential solver is ineffective.

10.2 A coefficient-pigeonhole construction

Fix pairwise distinct evaluation points $\alpha_1, \dots, \alpha_n \in \mathbb{F}_q$. For a subset $S \subseteq [n]$, write

$$V_S(X) := \prod_{i \in S} (X - \alpha_i). \quad (129)$$

The high coefficients of these monic vanishing polynomials determine a received word with a large agreement list.

Proposition 10.2 (Coefficient-pigeonhole bad ball). *Let $1 \leq k \leq A \leq n$. There is a received word $y \in \mathbb{F}_q^n$ for which*

$$|\mathcal{L}_{k-1}(y, A)| \geq \left\lceil \frac{\binom{n}{A}}{q^{A-k}} \right\rceil. \quad (130)$$

This holds for every choice of the distinct evaluation points.

Proof. For every A -element subset $S \subseteq [n]$, expand

$$V_S(X) = X^A + \sum_{j=0}^{A-1} v_{S,j} X^j \quad (131)$$

and assign to S the coefficient signature

$$\sigma(S) := (v_{S,k}, v_{S,k+1}, \dots, v_{S,A-1}) \in \mathbb{F}_q^{A-k}. \quad (132)$$

There are at most q^{A-k} signatures. Hence some signature σ is shared by a family C of at least the quantity on the right-hand side of Equation (130) distinct subsets.

Let

$$H_\sigma(X) := X^A + \sum_{j=k}^{A-1} \sigma_j X^j, \quad y_i := H_\sigma(\alpha_i). \quad (133)$$

For every $S \in C$, define

$$P_S(X) := H_\sigma(X) - V_S(X). \quad (134)$$

The coefficients of degrees $k, k+1, \dots, A$ cancel, so $\deg P_S < k$. Moreover,

$$P_S(\alpha_i) = y_i \iff V_S(\alpha_i) = 0 \iff i \in S.$$

Thus P_S agrees with y in exactly the A positions in S , and $P_S \in \mathcal{L}_{k-1}(y, A)$.

It remains to count distinct polynomials rather than subsets. If $P_S = P_T$, then Equation (134) gives $V_S = V_T$. The two polynomials are monic and their roots among the distinct α_i 's are exactly the elements indexed by their respective subsets, so $S = T$. The map $S \mapsto P_S$ is injective on C , which proves the proposition. $\square$

The same argument gives an exact description whenever the center is the evaluation of a monic degree- A polynomial whose coefficients of degrees at least k have been fixed.

Proposition 10.3 (Coefficient-fiber description). *Let $k \leq A \leq n$, and let*

$$F(X) = X^A + \sum_{j=k}^{A-1} c_j X^j.$$

For the received word $y_i = F(\alpha_i)$, the degree- $< k$ list at agreement A is in bijection with the A -element subsets $S \subseteq [n]$ for which the monic root polynomial $V_S = \prod_{i \in S} (X - \alpha_i)$ has coefficient c_j in every degree $k \leq j < A$. Every listed polynomial has exactly A agreements.

Proof. For such a subset, $P_S = F - V_S$ has degree less than k , and its agreement set is exactly S . Conversely, if a degree- $< k$ polynomial P has at least A agreements, then $F - P$ is monic of degree A and has at least A distinct roots among the evaluation points. It has exactly those roots, so it equals their monic root polynomial V_S . The coefficients in degrees at least k are therefore the prescribed c_j , and the two maps are inverse. $\square$

Corollary 10.4 (Two exact coefficient fibers). *Let p be prime.*

(i) *On the full evaluation set $\mathbb{F}_p$, let $2 \leq A \leq p-1$, $k = A-1$, and $y_x = x^A$. The list at agreement A has exactly*

$$\frac{1}{p} \binom{p}{A}$$

members.

(ii) *On $\mathbb{F}_p^\times$, let $A = 2r \leq p-1$, $k = 2r-1$, and $y_x = x^{2r}$. The list at agreement $2r$ has exactly*

$$\frac{\binom{p-1}{2r} + p-1}{p}$$

members.

Proof. By [Proposition 10.3](#), both lists count subsets whose elements sum to zero. For part (i), translation by $c \in \mathbb{F}_p$ changes the sum of an A -subset by Ac . Since $A \neq 0$ in $\mathbb{F}_p$, the p sum fibers are equinumerous.

For part (ii), let N_a be the number of zero-sum a -subsets of $\mathbb{F}_p^\times$. Splitting a zero-sum a -subset of $\mathbb{F}_p$ according to whether it contains zero gives

$$N_a + N_{a-1} = \frac{1}{p} \binom{p}{a}, \quad N_0 = 1.$$

Pascal's identity solves this recurrence as

$$N_a = \frac{\binom{p-1}{a} + (p-1)(-1)^a}{p}.$$

Substitute the even integer $a = 2r$. □

10.3 The exact capacity boundary

At agreement $A = k$, no pigeonhole loss occurs. In fact, the construction has exactly one nearby message for every k -element subset of coordinates.

Proposition 10.5 (Exact maximum list at capacity). *Let $1 \leq k \leq n$, and set*

$$y^{\text{cap}} := (\alpha_1^k, \dots, \alpha_n^k) \in \mathbb{F}_q^n. \quad (135)$$

Then

$$\max_{y \in \mathbb{F}_q^n} |\mathcal{L}_{k-1}(y, k)| = |\mathcal{L}_{k-1}(y^{\text{cap}}, k)| = \binom{n}{k}. \quad (136)$$

Proof. For every k -element subset $S \subseteq [n]$, the polynomial

$$P_S(X) := X^k - V_S(X) \quad (137)$$

has degree less than k and agrees with y^{cap} on every position in S . Distinct subsets give distinct polynomials by the injectivity argument in the proof of [Proposition 10.2](#).

Conversely, let P have degree less than k and agree with y^{cap} in at least k positions. The monic polynomial $X^k - P(X)$ has degree k , and the agreement positions are roots of this polynomial. It has at most k distinct roots, so it has exactly k of the evaluation points as roots. If S indexes these roots, monicity gives $X^k - P = V_S$, and hence $P = P_S$. The displayed family therefore exhausts the list.

The universal upper bound is (126) with $A = k$, which proves the maximum identity. □

For a fixed rate $R \in (0, 1)$ and $k/n \rightarrow R$, [Equation \(136\)](#) is $\exp((h(R) + o(1))n)$, where h is the binary entropy function in natural units. Thus no polynomial list-size bound is possible at radius $1 - R$, even when the evaluation set is fixed in advance.

10.4 A shrinking-gap obstruction

The coefficient-pigeonhole loss is exact enough to locate an obstruction inside a vanishing neighborhood of capacity. Define

$$h(x) := -x \log x - (1-x) \log(1-x), \quad 0 \log 0 := 0. \quad (138)$$

Proposition 10.6 (Entropy lower bound for a shrinking gap). *For integers $1 \leq k \leq k+s \leq n$,*

$$\log \max_{y \in \mathbb{F}_q^n} |\mathcal{L}_{k-1}(y, k+s)| \geq nh \left(\frac{k+s}{n} \right) - s \log q - \log(n+1). \quad (139)$$

Consequently, consider any sequence of evaluation sets over fields $\mathbb{F}_{q_n}$, with message dimensions k_n and nonnegative gaps s_n , such that $k_n + s_n \leq n$ and

$$\frac{k_n}{n} \longrightarrow R \in (0, 1), \quad \gamma_n := \frac{s_n}{n} \longrightarrow 0, \quad \limsup_{n \rightarrow \infty} \gamma_n \log q_n \leq c. \quad (140)$$

Then the worst-case list size L_n at agreement $k_n + s_n$ satisfies

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log L_n \geq \max\{0, h(R) - c\}. \quad (141)$$

In particular, if $\gamma_n \log q_n = o(1)$, then $L_n \geq \exp((h(R) - o(1))n)$.

Proof. Apply [Proposition 10.2](#) with $A = k+s$. For $a = (k+s)/n$, the standard type bound gives

$$\binom{n}{k+s} \geq \frac{\exp(nh(a))}{n+1}. \quad (142)$$

For completeness, when $0 < a < 1$, the $(k+s)$ -th summand is a mode of the binomial expansion $1 = (a + (1-a))^n$. Since there are at most $n+1$ summands, this mode is at least $1/(n+1)$. Dividing it by $a^{k+s}(1-a)^{n-k-s} = \exp(-nh(a))$ proves [Equation \(142\)](#); the endpoint cases follow directly. Combining this estimate with [Equation \(130\)](#) and taking logarithms proves [Equation \(139\)](#).

Under [Equation \(140\)](#), continuity of h gives $h(k_n/n + \gamma_n) \rightarrow h(R)$. Divide [Equation \(139\)](#) by n , take the lower limit, and use $L_n \geq 1$ to obtain [Equation \(141\)](#). $\square$

The last proposition gives a necessary scale, not only an endpoint example. If the worst-case lists are subexponential and $k_n/n \rightarrow R \in (0, 1)$, then any vanishing agreement gap must satisfy

$$\frac{s_n}{n} \log q_n \geq h(R) - o(1), \quad \text{or equivalently} \quad s_n \geq (h(R) - o(1)) \frac{n}{\log q_n}. \quad (143)$$

For fields of size $q_n = \Theta(n)$, a gap of $o(1/\log n)$ above rate still leaves $\exp((h(R) - o(1))n)$ codewords in one Hamming ball. This includes gaps of order $1/\log^2 n$ and n^{-c} for every fixed $c > 0$. The obstruction does not rule out polynomial lists once the gap reaches the scale $h(R)/\log q_n$; it identifies the strongest conclusion supplied by this construction.

10.5 Dependence of the list size on the gap

The same construction gives a much larger inverse-gap obstruction when the block length is chosen on the scale $2^{1/\delta}$.

Theorem 10.7 (A doubly exponential inverse-gap bad ball). *For every $0 < \delta \leq 1/16$, there is a prime $q = n$, a dimension k with*

$$\left| \frac{k}{n} - \frac{1}{2} \right| \leq \frac{\delta}{16},$$

and a received word $y \in \mathbb{F}_q^n$ such that, for $A = k + \lceil \delta n \rceil$,

$$|\mathcal{L}_{k-1}(y, A)| \geq \exp\left(\frac{\delta}{16} 2^{1/\delta}\right). \quad (144)$$

The center y is existential, obtained from the coefficient-pigeonhole construction.

Proof. Put $T = 2^{1/\delta}$ and $N = \lceil T/16 \rceil$. Bertrand's postulate gives a prime q with $N \leq q < 2N$, so

$$\frac{T}{16} \leq n = q \leq \frac{T}{4}. \quad (145)$$

For $\delta \leq 1/16$, these inequalities also give $q \geq 8/\delta$. Take $k = \lfloor q/2 \rfloor$, put $s = \lceil \delta q \rceil$, and set $A = k + s$. Since q is odd,

$$x := \frac{A}{q} - \frac{1}{2} = \frac{s - 1/2}{q}$$

satisfies $0 < x \leq 17\delta/16 < 1/4$. The binary entropy function from (138) has $|h''(1/2 + x)| \leq 16/3$ for $|x| \leq 1/4$. Taylor's theorem therefore gives

$$h(A/q) \geq \log 2 - \frac{8}{3}x^2 \geq \log 2 - 4\delta^2.$$

Also $s \leq \delta q + 1$, and $\log q \leq (\log 2)/\delta - \log 4$. Applying [Proposition 10.6](#) yields

$$\begin{aligned} \log \max_y |\mathcal{L}_{k-1}(y, A)| &\geq qh(A/q) - s \log q - \log(q + 1) \\ &\geq \delta q(\log 4 - 4\delta) - 2 \log q - 1. \end{aligned} \quad (146)$$

The inequality $\delta^2 T \geq 256 > 256 \log 2 + 128\delta$, valid throughout $0 < \delta \leq 1/16$, implies $2 \log q + 1 \leq \delta q/8$. Hence

$$\log \max_y |\mathcal{L}_{k-1}(y, A)| \geq \delta q(\log 4 - 1/4 - 1/8) > \delta q \geq \frac{\delta T}{16}.$$

Exponentiating proves (144). Finally,

$$\left| \frac{k}{q} - \frac{1}{2} \right| = \frac{1}{2q} \leq \frac{\delta}{16}.$$

□

Remark 10.8 (Quantifiers in the lower bound). The lengths in [Theorem 10.7](#) satisfy $n = \Theta(2^{1/\delta})$. The theorem does not give this list size at every sufficiently large length for a fixed gap, and these lengths may lie below the threshold N_δ in [Theorem 9.16](#) of the main paper. An unrestricted δ -dependent prefactor in an eventual list bound can absorb the example. Thus the theorem rules out all-length bounds polynomial in n and $1/\delta$, but it does not by itself lower-bound the derivative order in [Theorem 9.16](#) of the main paper.

The next construction, developed in this work, makes the inverse-gap obstruction explicit. It uses the multiplicative cosets of a prime field rather than pigeonholing coefficient signatures.

Theorem 10.9 (Explicit multiplicative-coset lists). *For every $0 < \delta \leq 1/8$, let*

$$t = \left\lceil \frac{1}{\delta} \right\rceil - 1.$$

There are infinitely many prime-field Reed–Solomon codes with $q = n + 1$, evaluation set $\mathbb{F}_q^\times$, and rate in $[1/5, 1/2]$, together with a received word whose list at agreement threshold $k + \lceil \delta n \rceil$ has at least

$$\frac{2^t}{t+1} \quad (147)$$

members.

Proof. Put $r = \lfloor t/2 \rfloor$. The definition of t gives $\delta t < 1$. By Dirichlet’s theorem on primes in arithmetic progressions [Dir37], infinitely many primes satisfy $q \equiv 1 \pmod{t}$. Choose one large enough that

$$h = \frac{q-1}{t} \geq \max\{2, (1-\delta t)^{-1}\},$$

and put $n = q - 1 = th$. The cyclic group $\mathbb{F}_q^\times$ has a subgroup of order h ; its t cosets are the root sets of

$$X^h - \beta_1, \dots, X^h - \beta_t$$

for the t distinct h -th powers β_i .

For an r -element subset $S \subseteq [t]$, define

$$G_S(X) = \prod_{i \in S} (X^h - \beta_i), \quad P_S(X) = X^{rh} - G_S(X).$$

The leading terms cancel, so $\deg P_S \leq (r-1)h$. Set $k = (r-1)h + 1$ and take the received word $y_\alpha = \alpha^{rh}$ on $\mathbb{F}_q^\times$. Then $P_S(\alpha) = y_\alpha$ precisely on the r selected cosets, giving exactly rh agreements. Moreover,

$$\delta n = \delta th \leq h - 1,$$

so $\lceil \delta n \rceil \leq h - 1$, and hence $k + \lceil \delta n \rceil \leq rh$. Distinct subsets give distinct monic polynomials G_S , and therefore distinct messages P_S . Finally,

$$\binom{t}{r} \geq \frac{2^t}{t+1}$$

because the largest of the $t+1$ binomial coefficients is at least their average. For $t \geq 4$ and sufficiently large h , the rate $k/n = (r-1)/t + 1/(th)$ lies in the asserted interval. $\square$

Since $t = \delta^{-1} + O(1)$, this gives the explicit bound

$$\frac{2^t}{t+1} = \exp\left(\frac{\log 2}{\delta} - O(\log(1/\delta))\right)$$

at infinitely many lengths, complementing the larger existential bound in [Theorem 10.7](#).

Corollary 10.10 (Exponential dependence on the inverse gap). *There are absolute constants $c, \delta_0 > 0$ such that, for every $0 < \delta < \delta_0$, some prime-field Reed–Solomon code of rate R satisfying $|R - 1/2| \leq \delta$ has a received word with at least*

$$\exp(c/\delta)$$

degree- $< k$ polynomials agreeing in at least $k + \lceil \delta n \rceil$ positions. In particular, no uniform worst-case list bound $\text{poly}(1/\delta)$ is possible for arbitrary prime-field evaluation sets.

Proof. Let $N = \lceil 1/\delta \rceil$. Bertrand’s postulate gives a prime p with $N \leq p < 2N$. Set $q = n = p$, use all elements of $\mathbb{F}_p$ as evaluation points, and take $k = \lfloor p/2 \rfloor$. Put $s = \lceil \delta p \rceil$ and $A = k + s$. Since $\delta p < 2\delta(1/\delta + 1)$, one has $s \leq 3$ for all sufficiently small δ , and $A \leq n$. By [Proposition 10.2](#),

$$\max_y |\mathcal{L}_{k-1}(y, A)| \geq \frac{\binom{p}{k+s}}{p^s}.$$

The ratio $(k + s)/p$ lies in $[2/5, 3/5]$ for large p . The type bound (142) therefore gives

$$\log \max_y |\mathcal{L}_{k-1}(y, A)| \geq p \min_{x \in [2/5, 3/5]} h(x) - 3 \log p - \log(p + 1).$$

For sufficiently large p , this is at least $c_0 p$ for an absolute $c_0 > 0$. Since $p \geq 1/\delta$, the claim follows after decreasing the constant. $\square$

11 Refreshed application certificate values

The application checkers distributed with this companion now select the tight first-order Taylor-degree model explicitly. The certificate engine’s default remains the legacy model, so historical commands and stored baselines retain their original meaning.

ProveKit over cubic Goldilocks. For the second code in the retuned lookup configuration, with $(n, k, A) = (2048, 128, 453)$ and support $(38, 23, 133)$, the squarefree list bound is

$$\Lambda \leq \frac{2659384185}{326}.$$

The machine-readable certificate records the safe generic ceiling 8157621. Since Λ is an integer, the separation calculation can instead use the sharper floor 8157620, which gives 137.0819759979 bits with one out-of-domain evaluation. The successive-stage alternative has exact bound 15314557817/163 and ceiling 93954343, giving 130.0304965539 bits in the same calculation. These are analytical bounds for the fixed retuned configuration, not measured proof sizes.

ZisK compressed final proof. At 51 queries and 22 grinding bits, the two batching stages contribute the exact error

$$\frac{4073087021134687044}{(2^{64} - 2^{32} + 1)^3},$$

or 130.1791716533 bits. The corresponding query term has 128.0003360286 bits under the checker’s independent-uniform-output model. The native saved proofs remain 286013 and 272783 bytes. By contrast, the analytical field-vector accounting is 254032 and 242272 bytes; the latter figures are not serialized proof-file measurements.

LambdaVM CPU certificate. For the list that selects the 38 main-column polynomials after the two early evaluations, the tight certificate gives

$$\Lambda \leq \frac{367473974837}{3230}, \quad \lceil \Lambda \rceil = 113769033.$$

With the existing cubic-Goldilocks field, 20 grinding bits, 208 CPU queries, and the unchanged non-CPU profiles, the complete changed-component bound has 128.0627324732 bits. The analytical field-and-hash accounting saves 55992 bytes. The measured rkyv reduction below is 59832 bytes because it includes the serializer’s actual query records, metadata, alignment, and the two new evaluation vectors.

12 Measured LambdaVM proof savings and overhead

The first-order bounds also support a concrete change to LambdaVM’s CPU proof: reduce its query count from 219 to 208 and add two early evaluations of its 38 main columns. We implement this experimental profile on a fork. On the same 32768-instruction assembly benchmark and empty input, the complete CPU subproof shrinks from 1 233 024 to 1 173 192 serialized bytes: a saving of 59 832 bytes (59.832 decimal KB, 4.85%) [Dao26a]. This comparison includes every CPU commitment, claim, and opening, including the added evaluations, and excludes shared VM-level metadata. All 25 non-CPU subproofs keep 219 queries. The complete VM proof decreases by the same 59 832 bytes, from 36 868 888 to 36 809 056. Both proofs are generated, deserialized, and verified through the release CLI with their matching, caller-selected profiles. This is one measured workload, not an average over programs.

Separating the two changes. To identify the cost of the added evaluations, we cross their presence with the CPU query count. The resulting four cells have the following complete VM-proof sizes under rkyv serialization.

Cell	CPU queries	Early evaluations	Bytes
Control	219	0	36 868 888
Evaluations only	219	2	36 870 744
Queries only	208	0	36 807 200
Actual profile	208	2	36 809 056

The two early vectors add 1856 bytes; removing eleven query records saves 61 688 bytes. These effects are additive, giving the 59 832-byte net saving. They include the serializer’s metadata and alignment. The middle two cells are measurement instruments, not proposed protocol profiles; in particular, the 208-query cell without early evaluations is not certified by the paper’s argument.

End-to-end timing and added work. The normal-grinding experiment uses an Apple M4 Max with 16 threads, release builds, and grinding factor 20. We rotate the four cells through a balanced execution order and compare the actual profile with the control within each round. The reported 95% intervals use a round-level percentile bootstrap of the paired mean [Dao26b].

Measurement	Rounds per cell	Change (ms)	95% interval (ms)
Proving	20	−15.2	[−49.4, +18.9]
Whole verification	50	+0.02	[−0.62, +0.52]
CPU verifier components	50	−0.081	[−0.117, −0.048]

Neither end-to-end comparison resolves a latency change on this benchmark. The component measurements explain why verification can nonetheless do less work: at 208 queries, evaluations add 0.167 ms, whereas removing eleven queries with evaluations enabled saves 0.291 ms. These are conditional effects from the four-cell experiment, not the difference between the actual profile and control; that net difference is the −0.081 ms component entry above.

The prover’s extra arithmetic is visible in diagnostic runs with grinding disabled: the actual profile retires 264.4 million more instructions, about 0.1845% of the control. Three-round single-thread isolation attributes roughly 305 ms to the evaluation and DEEP work. This serial cost is not a 16-thread latency estimate, since the table proofs run concurrently. The largest new transient buffer is 3 MiB; process RSS, roughly 1.9 GiB, does not resolve it. Separate encode, decode, and archive access measurements differ by less than 0.05 ms. These codec runs were not interleaved, so their differences are descriptive.

Scope and reproduction. The experimental profile enforces one 32768-row CPU chunk and retains cubic Goldilocks, blowup two, and the baseline grinding and batching. The mathematical certificate controls the changed CPU component; neither the certificate nor these measurements establish a new full-VM security theorem. The implementation and complete-proof reproduction record are pinned in [Dao26a]. The supplied timing data and measurement report are distributed with this companion in `experiments/lambdavam-anchors`. Its checker verifies recorded byte arithmetic and recomputes paired mean timing contrasts; it does not rerun the prover or regenerate the reported bootstrap intervals.

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Disclosure

The second and third authors are Research Partners at a16z crypto, which is an investor in various blockchain-based platforms, as well as in the crypto ecosystem more broadly. General a16z disclosures are available at <https://www.a16z.com/disclosures/>.

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A Exact counting identities

We collect two elementary evaluations used in the finite and coarse interpolation estimates.

Lemma A.1 (Closed form for the staircase count). *For integers $D \geq 1$ and t , the function in (28) satisfies*

$$N_D(t) = 0 \quad \text{if } t \leq 0,$$

and, if $t > 0$, then

$$N_D(t) = (s+1)t - \frac{Ds(s+1)}{2}, \quad s = \left\lfloor \frac{t-1}{D} \right\rfloor. \quad (148)$$

Proof. When $t \leq 0$, no nonnegative pair (a, b) satisfies $a + Db < t$. Suppose $t > 0$. For a fixed b , the admissible values are $a = 0, \dots, t - Db - 1$, so there are $t - Db$ of them. Such a value exists exactly for $0 \leq b \leq s$. Summing the arithmetic progression gives (148). $\square$

We next evaluate the aggregate residual-rank count used in the enlarged-map analysis. This is the specialization of (43) with $M = m = d^3$ after replacing every $\Lambda_d(W + s)$ by its common upper bound. Define

$$h_s = \left\lceil \frac{m-s}{d} \right\rceil, \quad \beta_s = (s+1)(m+1) - (s-h_s+1)_+(m-h_s+1)_+.$$

Lemma A.2 (Exact aggregate residual-rank count). *Let $d \geq 2$ and $m = d^3$. Then*

$$\sum_{s=0}^{m-1} \beta_s = \frac{d^2(d^2-d+1)(2d^2+d+1)(4d^2-3d+5)}{12}. \quad (149)$$

Consequently,

$$\frac{1}{m^3} \sum_{s=0}^{m-1} \beta_s = \frac{(d^2 - d + 1)(2d^2 + d + 1)(4d^2 - 3d + 5)}{12d^7} < \frac{2}{3d}. \quad (150)$$

Proof. Parameterize s uniquely as

$$s = m - dh + t, \quad 1 \leq h \leq d^2, \quad 0 \leq t < d.$$

Then $h_s = h$. Moreover,

$$s - h + 1 = d^3 - (d + 1)h + t + 1.$$

The identity

$$d^3 + 1 = (d + 1)(d^2 - d + 1)$$

shows that this quantity is nonnegative exactly when $h \leq d^2 - d + 1$; at the last such value it equals t . Since $m - h + 1 > 0$, the positive part can therefore be removed after restricting the second sum to that range. Hence

$$\begin{aligned} \sum_{s=0}^{m-1} \beta_s &= (m + 1) \sum_{h=1}^{d^2} \sum_{t=0}^{d-1} (m - dh + t + 1) \\ &\quad - \sum_{h=1}^{d^2-d+1} \sum_{t=0}^{d-1} (m - (d + 1)h + t + 1)(m - h + 1). \end{aligned} \quad (151)$$

Using

$$\sum_{t=0}^{d-1} t = \frac{d(d-1)}{2}, \quad \sum_{h=1}^H h = \frac{H(H+1)}{2}, \quad \sum_{h=1}^H h^2 = \frac{H(H+1)(2H+1)}{6}$$

in (151), with $H = d^2 - d + 1$, gives (149) after collecting factors.

It remains to prove the strict inequality. After multiplying by $12d^7$, it is equivalent to

$$(d^2 - d + 1)(2d^2 + d + 1)(4d^2 - 3d + 5) < 8d^6.$$

The difference between the right and left sides is

$$d^4(10d - 21) + d^2(11d - 14) + 3d - 5.$$

It equals 17 at $d = 2$, and every displayed group is positive for $d \geq 3$. □

B Source-level repairs to the low-rate proof

This appendix records the points at which our reconstruction differs from the formal statements and notation of Brakensiek, Chen, Putterman, Zhang, and Zheng [BCP⁺26]. The hidden-derivative identity and the construction of the local kernel subspaces remain unchanged. We additionally prove that these subspaces exhaust the kernel of the enlarged map. The repairs to the published argument concern one quantifier in the global dimension comparison, one rounding step, and one rank-label slip.

B.1 The dimension quantifier in Proposition 3.13

In the source notation, Lemma 3.2 gives the dimension estimate

$$\dim Q > \frac{\theta^3}{384} |\mathcal{B}| (k-1) m^3, \quad (152)$$

while Lemma 3.12 bounds the rank imposed at one received point by

$$|\mathcal{B}| m^3 d^{-2\theta/(5+\theta)}. \quad (153)$$

The source proceeds by verifying the following sufficient inequality:

$$\frac{\theta^3}{384} \frac{k-1}{n} d^{2\theta/(5+\theta)} > 1. \quad (154)$$

This is the interface used between the source's dimension and rank bounds.

Proposition 3.13 assumes only $k/n \leq (1-\theta)\varepsilon$, but its proof then uses $k = \lfloor (1-\theta)\varepsilon n \rfloor$ to lower-bound $(k-1)/n$ [BCP⁺26, Section 3.3, Proposition 3.13]. An upper bound on k/n does not imply the lower bound required by Equation (154). The proof therefore establishes the proposition at the design dimension used inside the proof, not simultaneously at every smaller message dimension.

There are two direct repairs. The local statement can be restricted to $k = \lfloor (1-\theta)\varepsilon n \rfloor$, together with the finite-size condition needed for the asserted lower bound on $(k-1)/n$. To recover the claimed “rate at most” quantifier, run interpolation and root finding at an *ambient* dimension K satisfying

$$k \leq K < A, \quad \frac{K-1}{n} \geq \rho_* > 0, \quad (155)$$

and replace $k-1$ by $K-1$ throughout the interpolation support and the root-finding call. Every degree-less-than- k message is an ambient candidate, and candidates of degree at least k are removed after root finding. The source proof can take $K = \lfloor (1-\theta)\varepsilon n \rfloor$. For the one-sided support in this companion, the adaptive choice in Equation (60) supplies both inequalities in Equation (155) uniformly over the actual message dimension. The main paper uses a different ambient baseline in its weighted-support certificate (Theorem 6.8).

B.2 The ceiling in Lemma 3.2

The source support in equations (17) and (18) contains the condition

$$|c| \leq \left\lceil \left(1 + \frac{3\theta}{4}\right) m \right\rceil. \quad (156)$$

The proof of Lemma 3.2 replaces this ceiling by a floor and concludes $|c| \leq (1 + 3\theta/4)m$, which is not valid as written [BCP⁺26, Section 3.1, equations (17)–(19)]. The extra unit can be absorbed without changing the support or the final parameters.

Indeed, the rate assumption and $A = \lceil \varepsilon n \rceil$ give, for $k > 1$,

$$\frac{A}{k-1} > \frac{1}{1-\theta}. \quad (157)$$

Combining Equations (156) and (157) yields the exact slack estimate

$$\begin{aligned} \frac{mA}{k-1} - |c| &> \frac{m}{1-\theta} - \left(1 + \frac{3\theta}{4}\right) m - 1 \\ &= \frac{\theta m}{4} + \frac{\theta^2 m}{1-\theta} - 1. \end{aligned} \quad (158)$$

Consequently, the source's equation (19) follows whenever

$$\theta^2 m \geq 1 - \theta. \quad (159)$$

This condition is already forced by the source parameter regime. To see this, its equation (20) implies $\varepsilon < \theta^{15}$, because its base is smaller than θ^3 and its exponent is larger than five. Hence $d = \lceil \varepsilon^{-3/\theta} \rceil > \theta^{-45/\theta}$ and $m = d^3 > \theta^{-135/\theta}$, which gives $\theta^2 m > 1 > 1 - \theta$. Replacing the two incorrect rounding lines by [Equation \(158\)](#) repairs Lemma 3.2 with its original ceiling.

B.3 Rank notation

The source factors the local constraint map as $\Phi_{\alpha,y} = \Gamma \circ \Psi_{\alpha,y}$, and therefore $\text{rank}(\Phi_{\alpha,y}) \leq \text{rank}(\Gamma)$. In the proof of Lemma 3.12, rank-nullity and the kernel spaces from Lemma 3.11 are then applied to Γ , but the first displayed rank in that calculation is written as $\text{rank}(\Phi)$ [[BCP⁺26](#), Section 3.2.2, equations (21)–(24)]. The correct chain is

$$\text{rank}(\Phi_{\alpha,y}) \leq \text{rank}(\Gamma) \leq \dim \mathcal{V} - \dim \left(\bigoplus_{r=0}^{m-1} \mathcal{K}_r \right). \quad (160)$$

The remainder of the proof bounds the last expression, so this is a label correction rather than a change to the kernel argument. The corresponding factorization and rank calculation appear in [Equation \(38\)](#) and [Lemma 4.5](#). The source proves the kernel inclusion needed for this bound; the reverse inclusion and resulting exact formula for $\text{rank}(\Gamma)$ are new to this work.

B.4 Hypotheses at the interfaces

For later use, we make explicit four hypotheses that are implicit or only stated elsewhere in the source.

- (i) Proposition 3.13 must fix pairwise distinct evaluation points. Its final multiplicity argument counts A distinct roots, and this is exactly where distinctness is used.
- (ii) The source interpolation support contains $mA/(k-1)$, so its standalone statement requires $k \geq 2$. Lemma 3.12 uses $d \geq 3$ in its last numerical estimate. The later root-finding invocation additionally requires $k > d$.
- (iii) The local conditions in source equation (15) are imposed only for indices with $db < m$. Statements that quantify over every $b \geq 0$ must either retain this restriction or adopt the convention that no divisibility condition is imposed once $db \geq m$.
- (iv) The interpolation argument through Proposition 3.13 is characteristic-free. Primality of q , the inequalities $q \geq k > d$, the individual degree conditions $\deg_{Y_i} Q < q$, and the weighted-degree condition belong to the source root-finding theorem [[BCP⁺26](#), Theorem 2.1], not to the local interpolation lemma.

With these interfaces stated, the source equations (13)–(16) correspond to [Equations \(7\), \(11\) and \(12\)](#); the global multiplicity step of Proposition 3.13 corresponds to [Proposition 3.4](#). Our changes begin at the choice of ambient degree and interpolation support, not at the hidden-derivative local identity.

C A fixed-gap boundary asymptotic

The main paper uses a weighted support to obtain its explicit derivative order. For comparison, this appendix records a sharper asymptotic analysis of the one-sided support from [Section 4](#). It identifies a natural Gumbel boundary law and an exact one-dimensional variational constant. The result is modular and is not used in the proof of [Theorem 9.16](#) of the main paper. Its limit keeps the rate and agreement fixed while $d \rightarrow \infty$; it does not justify a joint limit as the gap tends to zero.

Recall the staircase function G from [Equation \(10\)](#). For an integer s and $D \geq 1$, the exact identity

$$N_D(s) = D G(s/D) \quad (161)$$

follows by summing, for each $j \geq 0$, the $s - Dj$ possible X -exponents. Fix constants $0 < \rho < \eta \leq 1$ and put

$$h = \frac{\eta}{\rho}, \quad \theta = 1 - \frac{1}{h}. \quad (162)$$

For fixed d, m, M, W , if $D/n \rightarrow \rho$ and $A/n \rightarrow \eta$, then

$$\frac{\dim Q}{n} \rightarrow \rho \mathcal{D}_d(h), \quad \mathcal{D}_d(h) := \sum_{\omega(c) \leq W} \sum_{b=0}^M G(mh - b - |c|). \quad (163)$$

This is an immediate application of [Equation \(161\)](#) to the exact dimension formula and continuity of G . Thus the limiting one-sided certificate is

$$\rho \mathcal{D}_d(h) > R_d(m, M, W). \quad (164)$$

For $\tau > 0$ and an integer $j \geq 0$, define

$$I_j(\tau) := \int_{\tau}^{\infty} \log^j(u/\tau) e^{-u} du, \quad C(h) := \min_{\tau > 0} \frac{6\tau^{-1/h}}{I_3(\tau)}. \quad (165)$$

Theorem C.1 (One-sided boundary ratio). *Fix $0 < \rho < \eta \leq 1$, $h = \eta/\rho$, and $\tau > 0$. Let $H = H_{d-1}$, let γ denote Euler's constant, and set*

$$\xi = -h(\gamma + \log \tau), \quad \alpha = h - \frac{\xi}{H}. \quad (166)$$

Choose

$$m = d^3, \quad M = \left\lfloor \frac{m}{\sqrt{H}} \right\rfloor, \quad W = \left\lfloor \frac{\alpha dm}{H} \right\rfloor. \quad (167)$$

Then, as $d \rightarrow \infty$,

$$\frac{\rho \mathcal{D}_d(h)}{R_d(m, M, W)} = (1 + o(1)) \frac{\eta I_3(\tau) \tau^{1/h}}{6} \frac{d^{\theta}}{H_{d-1}}. \quad (168)$$

The error is uniform when (ρ, η, τ) ranges over a compact subset of $\{0 < \rho < \eta \leq 1, \tau > 0\}$.

Proof. We give the probabilistic and endpoint estimates that determine every factor in [Equation \(168\)](#). Let

$$S_W = \left\{ x \in \mathbb{R}_{\geq 0}^{d-1} : \sum_{i=1}^{d-1} ix_i \leq W \right\}, \quad V_W = \frac{W^{d-1}}{((d-1)!)^2}.$$

If X is uniform on S_W and $E_0, \dots, E_{d-1}$ are independent unit exponential random variables, then the standard exponential representation of the uniform simplex gives

$$X_i = \frac{WE_i}{i \sum_{j=0}^{d-1} E_j}. \quad (169)$$

Consequently, for

$$T_d = \frac{d}{\sum_{j=0}^{d-1} E_j} \sum_{i=1}^{d-1} \frac{E_i}{i},$$

we have $|X|/m = (\alpha/H + o(1))T_d$. More precisely, the definition of W gives

$$\frac{W}{dm} = \frac{\alpha}{H} + O\left(\frac{1}{dm}\right),$$

so the coefficient error is negligible on the boundary scale used below.

The sum $\sum_{i=1}^{d-1} E_i/i$ has the same distribution as the maximum of $d-1$ independent unit exponentials. Hence

$$T_d - H \implies Z := G_{\text{Gum}} - \gamma, \quad (170)$$

where $\Pr(G_{\text{Gum}} \leq z) = \exp(-e^{-z})$. The convergence holds with uniformly integrable cubes. Indeed, the maximum M_N satisfies

$$\Pr(M_N - \log N > t) \leq e^{-t}, \quad \Pr(M_N - \log N < -t) \leq e^{-e^t},$$

and the factor $d/\sum E_j$ differs from one in L^6 by $O(d^{-1/2})$. It follows from [Equation \(166\)](#) that

$$H\left(h - \frac{|X|}{m}\right) \implies \xi - hZ \quad (171)$$

with uniformly integrable positive cubes.

Now put $a_m = m/H$ and

$$F_{m,M}(t) := \sum_{b=0}^M \mathbf{G}(t-b).$$

Since $M/a_m = \sqrt{H} + o(1)$, comparison with a Riemann integral gives, uniformly on compact x -sets,

$$a_m^{-3} F_{m,M}(a_m x) \longrightarrow \frac{x_+^3}{6}. \quad (172)$$

Indeed, the elementary bounds

$$\frac{x_+^2}{2} \leq \mathbf{G}(x) \leq \frac{x_+^2}{2} + x_+$$

reduce the normalized sum to a Riemann sum for $\frac{1}{2} \int_0^\infty (x-t)_+^2 dt$. They also give the uniform envelope $a_m^{-3} F_{m,M}(a_m x) \leq C(1 + x_+^3)$.

We next justify the lattice transfer, including the floors. Put $n_d = d-1$, $B_d = \binom{d}{2}$, and

$$U_W := \bigcup_{\omega(c) \leq W} (c + [0, 1)^{n_d}).$$

Then $S_W \subseteq U_W \subseteq S_{W+B_d}$. Since $F_{m,M}$ is nondecreasing and $|c| \leq |x| < |c| + n_d$ on the cube based at c , we have the exact sandwich

$$\begin{aligned} \int_{S_W} F_{m,M}(mh - |x|) dx &\leq \mathcal{D}_d(h) \\ &\leq \int_{S_{W+B_d}} F_{m,M}(mh - |x| + n_d) dx. \end{aligned} \quad (173)$$

Here

$$\frac{B_d}{W} = O\left(\frac{H}{d^2}\right), \quad \frac{n_d H}{m} = O\left(\frac{H}{d^2}\right), \quad \frac{\text{Vol}(S_{W+B_d})}{V_W} = 1 + O\left(\frac{H}{d}\right) = 1 + o(1).$$

Increasing the simplex radius by B_d perturbs the normalized boundary slack by $O_{L^3}(dH^2/m) = o(1)$, using the exponential representation and the cubic moment bound above. The additive cube shift perturbs it by $O(dH/m) = o(1)$. The envelope following Equation (172), together with the uniform integrability in Equation (171), therefore shows that the two normalized integrals in Equation (173) have the same limit.

If U is unit exponential, then $-\log U$ is standard Gumbel, and

$$\xi - hZ = h \log(U/\tau).$$

Thus Equations (171) and (172) yield

$$\frac{\mathcal{D}_d(h)}{V_W} = (1 + o(1)) \frac{m^3 h^3}{6H^3} I_3(\tau). \quad (174)$$

It remains to evaluate the exact enlarged-map rank. Write its index as $r = m - s$ and put $u_s = \lceil s/d \rceil$. The unit-cube bounds give, uniformly for $1 \leq s \leq m$,

$$\left(\frac{W + m - s}{W}\right)^{d-1} \leq \frac{\Lambda_d(W + m - s)}{V_W} \leq \left(\frac{W + m - s + B_d}{W}\right)^{d-1}. \quad (175)$$

Since $d(m/W)^2 = O(H^2/d)$ and $dB_d/W = O(H/d)$, expanding the logarithms in Equation (175) yields the uniform estimate

$$\frac{\Lambda_d(W + m - s)}{V_W} = \exp\left(\frac{H}{\alpha} - \frac{Hs}{\alpha m} + o(1)\right). \quad (176)$$

The exact residual factor from Equation (43) is

$$\beta_s := (m - s + 1)(M + 1) - (m - s - u_s + 1)_+ (M - u_s + 1)_+.$$

For fixed $0 < \varepsilon < L$, uniformly on $\varepsilon m/H \leq s \leq Lm/H$, both positive parts are active and direct expansion gives

$$\beta_s = (1 + o(1)) \frac{ms}{d}. \quad (177)$$

For every $1 \leq s \leq m$, the exact definition also gives

$$\beta_s \leq \left(\frac{s}{d} + 1\right) (m + M + 2). \quad (178)$$

To see this when both positive parts are active, use $\beta_s = u_s(m - s + M + 2) - u_s^2$; if one is inactive, the same upper bound follows from $u_s > m - s + 1$ or $u_s > M + 1$.

Set $\lambda = H/(\alpha m)$. Insert Equations (176) and (177) in the rank sum first on $\varepsilon m/H \leq s \leq Lm/H$. The global bound Equation (178) controls the complementary ranges. After normalization by $m^3/(dH^2)$, the range $s < \varepsilon m/H$ contributes $O(\varepsilon^2)$, while the range $s > Lm/H$ contributes $O((L + 1)e^{-L/(2\alpha)})$. The terms arising from the $+1$ and M in Equation (178) are $o(1)$ because $dH/m = o(1)$ and $M/m = o(1)$. Letting first $d \rightarrow \infty$, then $\varepsilon \downarrow 0$ and $L \rightarrow \infty$, reduces the rank to the geometric sum

$$\sum_{s=1}^m s e^{-\lambda s} = (1 + o(1)) \lambda^{-2} = (1 + o(1)) \frac{\alpha^2 m^2}{H^2}.$$

Here the first equality follows from $\sum_{s \geq 1} sz^s = z/(1-z)^2$ with $z = e^{-\lambda}$; truncation after m has relative error $O((1 + \lambda m)e^{-\lambda m}) = O(Hd^{-1/\alpha+o(1)}) = o(1)$. Consequently,

$$\frac{R_d(m, M, W)}{V_W} = (1 + o(1)) \frac{m^3 \alpha^2 e^{H/\alpha}}{dH^2}. \quad (179)$$

Divide Equation (174) by Equation (179) and multiply by ρ . Since $H = \log d + \gamma + o(1)$, $\alpha = h - \xi/H$, and $\rho h = \eta$,

$$\frac{d}{e^{H/\alpha}} = (1 + o(1)) d^{1-1/h} \tau^{1/h}.$$

Substitution proves Equation (168). Every comparison used above is uniform on the stated compact sets. $\square$

Lemma C.2 (The variational constant). *For every $h \geq 1$, the minimum defining $C(h)$ exists and has a unique minimizer $\tau_h \in (0, 1/h)$, characterized by*

$$I_3(\tau_h) = 3hI_2(\tau_h). \quad (180)$$

Moreover, $C(h)$ is strictly decreasing in h .

Proof. The objective tends to infinity as $\tau \downarrow 0$, since $I_3(\tau) = O((1 + |\log \tau|)^3)$, and also as $\tau \rightarrow \infty$, since $I_3(\tau) \leq 6e^{-\tau}/\tau^3$. It therefore has an interior minimizer. For $j \geq 1$, differentiation under the integral gives $I'_j(\tau) = -(j/\tau)I_{j-1}(\tau)$, so every stationary point satisfies Equation (180).

To prove uniqueness, put

$$r(\tau) := \frac{I_3(\tau)}{3I_2(\tau)}.$$

After the change of variables $u = \tau v$, this is one third of the expectation of $\log V$ under the probability density on $v > 1$ proportional to $\log^2(v)e^{-\tau v}$. Differentiating this expectation with respect to τ gives

$$r'(\tau) = -\frac{1}{3} \text{Cov}_\tau(\log V, V) < 0,$$

because both functions are strictly increasing and the distribution is nondegenerate. As $\tau \rightarrow \infty$, integration by parts gives $I_3/I_2 < 3/\tau \rightarrow 0$. As $\tau \downarrow 0$, restricting the integral for I_3 to $u \in [1, 2]$ gives $I_3 = \Omega(\log^3(1/\tau))$, while direct expansion of $\log(u/\tau)$ gives $I_2 = O(\log^2(1/\tau))$. Hence $r(\tau) \rightarrow \infty$ at zero and $r(\tau) \rightarrow 0$ at infinity. Thus $r(\tau) = h$ has exactly one solution, and the logarithmic derivative of the objective changes from negative to positive there. This solution is the unique minimizer. Integration by parts gives $I_3(\tau) < 3I_2(\tau)/\tau$, hence $\tau_h < 1/h$.

Finally, if $h_2 > h_1$, evaluate the h_2 -objective at $\tau_{h_1} < 1$. The factor $\tau^{-1/h}$ strictly decreases with h at that point, so $C(h_2) < C(h_1)$. $\square$

Corollary C.3 (Fixed-gap one-sided certificate). *Fix $\delta \in (0, 1)$ and $\zeta > 0$, and put*

$$\rho_0 = \frac{\delta(1-\delta)}{2}, \quad h_0 = (1-\delta)^{-1}.$$

Choose a minimizer τ_0 for $C(h_0)$. There is $d_0 = d_0(\delta, \zeta)$ with the following property. Let $d \geq d_0$ be an integer satisfying

$$\frac{d^\delta}{H_{d-1}} > (1 + \zeta)C((1-\delta)^{-1}), \quad (181)$$

and set $H = H_{d-1}$, $m = d^3$, and $M = \lfloor m/\sqrt{H} \rfloor$. There is $N = N(\delta, \zeta, d)$ such that the following holds for every $n \geq N$ and $1 \leq k \leq n$. Set $A := k + \lceil \delta n \rceil$. If $A \leq n$, define

$$\begin{aligned} K &:= \max\{k, \lceil \rho_0 n \rceil\}, & D &:= K - 1, & h_{n,k} &:= \frac{A}{D}, \\ \xi_{n,k} &:= -h_{n,k}(\gamma + \log \tau_0), & \alpha_{n,k} &:= h_{n,k} - \frac{\xi_{n,k}}{H}, \\ W_{n,k} &:= \left\lfloor \frac{\alpha_{n,k} dm}{H} \right\rfloor. \end{aligned}$$

Then $d < D$, and the exact finite inequality [Equation \(58\)](#) holds with parameters $(D, A, d, m, M, W_{n,k})$. Thus the derivative order is rate-independent, while the support threshold $W_{n,k}$ is explicitly rate-adaptive.

Proof. For orientation, first associate to an actual rate $x \in [0, 1 - \delta]$ the limiting quantities

$$\rho_x = \max\{x, \rho_0\}, \quad \eta_x = x + \delta, \quad h_x = \eta_x / \rho_x, \quad \theta_x = 1 - 1/h_x.$$

Then $h_x \geq h_0$. If

$$C_{\tau_0}(h) := \frac{6\tau_0^{-1/h}}{I_3(\tau_0)},$$

the proof of [Lemma C.2](#) gives $C_{\tau_0}(h_x) \leq C(h_0)$.

Suppose first that $x \geq \rho_0$. Then $\theta_x = \delta/\eta_x$. Once $\delta \log d \geq 1$,

$$\frac{d}{d\eta} \log(\eta d^{\delta/\eta - \delta}) = \frac{\eta - \delta \log d}{\eta^2} \leq 0 \quad (\delta \leq \eta \leq 1).$$

The expression inside the logarithm is one at $\eta = 1$, so

$$\eta_x d^{\theta_x - \delta} \geq 1. \tag{182}$$

If $x < \rho_0$, then $\eta_x \geq \delta$ and $\theta_x \geq (1 + \delta)/2$. For all sufficiently large d ,

$$\eta_x d^{\theta_x - \delta} \geq \delta d^{(1-\delta)/2} \geq 1. \tag{183}$$

Combining [Equation \(181\)](#) with [Equations \(182\)](#) and [\(183\)](#) gives, uniformly in x ,

$$\frac{\eta_x d^{\theta_x}}{H} > (1 + \zeta) C_{\tau_0}(h_x).$$

By [Theorem C.1](#), the left-to-right ratio in the limiting certificate is therefore larger than one with a uniform positive margin, provided d_0 is sufficiently large. The parameters (ρ_x, η_x, h_x) range over a compact set with $\rho_x \geq \rho_0$, so the uniformity assertion in that theorem applies.

We now apply this comparison directly to each finite instance, avoiding any continuity assertion across the floor in $W_{n,k}$. Write

$$\widehat{\rho} = \frac{D}{n}, \quad \widehat{\eta} = \frac{A}{n}, \quad \widehat{h} = \frac{A}{D}, \quad \widehat{\theta} = 1 - \frac{1}{\widehat{h}}, \quad x = \frac{k}{n}.$$

For every sufficiently large n , $\widehat{\rho} < \rho_x$ and $\widehat{\eta} \geq \eta_x$. Hence $\widehat{h} \geq h_x$, $\widehat{\theta} \geq \theta_x$, and

$$\widehat{\eta} d^{\widehat{\theta} - \delta} \geq \eta_x d^{\theta_x - \delta} \geq 1.$$

Also $C_{\tau_0}(\widehat{h}) \leq C_{\tau_0}(h_x)$. It follows that

$$\frac{\widehat{\eta} d^{\widehat{\theta}}}{H} > (1 + \zeta) C_{\tau_0}(\widehat{h}). \quad (184)$$

The triples $(\widehat{\rho}, \widehat{\eta}, \tau_0)$ lie in one compact subset of $\{0 < \rho < \eta \leq 1, \tau > 0\}$ for all large n . The uniform version of [Theorem C.1](#), applied with the actual floored $W_{n,k}$, therefore turns [Equation \(184\)](#) into

$$\widehat{\rho} \mathcal{D}_d(\widehat{h}) > \mathbf{R}_d(m, M, W_{n,k})$$

with a uniform positive margin. Choose d_0 large enough also to ensure $\alpha_{n,k} > 0$ throughout this compact set.

Finally, the exact staircase identity gives

$$\begin{aligned} \frac{\dim Q(D, A, d, m, M, W_{n,k})}{n} &= \widehat{\rho} \sum_{\omega(c) \leq W_{n,k}} \sum_{b=0}^M \\ &\quad \cdot G \left(m\widehat{h} - (1 - D^{-1})b - \sum_{j=2}^d (1 - jD^{-1})c_j \right). \end{aligned}$$

The argument of G in each exact summand is

$$m\widehat{h} - b - |c| + \frac{b + \sum_{j=2}^d j c_j}{D},$$

which is at least $m\widehat{h} - b - |c|$. Since $\mathbf{G}$ is nondecreasing, the exact dimension is therefore at least $\widehat{\rho} \mathcal{D}_d(\widehat{h})$, with no limiting error. The strict comparison above proves the finite certificate. Increasing N if necessary also gives $d < D$. $\square$

The corollary identifies the sharp leading constant of this particular one-sided enlarged-map certificate at its worst rate. It does not show that the actual local rank, another interpolation support, or the weighted support cannot do better. In fact, the weighted support is what gives the stronger explicit $\exp(1.5/\delta)$ derivative order in the main paper.